

MINISTRY OF EDUCATION AND TRAINING THE STATE BANK OF VIETNAM

HO CHI MINH UNIVERSITY OF BANKING



BACHELOR THESIS

**SENTIMENT-ENHANCED MACHINE LEARNING FOR
CORPORATE DEFAULT PREDICTION: A TREE-BASED
APPROACH WITH MONOTONIC CONSTRAINTS**

MAJOR: FINANCE AND BANKING

CODE: 7 34 02 01

NGO HOANG KHANH DUY

HO CHI MINH CITY, 2025

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ABSTRACT

This graduation thesis examines how financial ratios and news-based sentiment jointly inform corporate default prediction in Vietnamese listed firms, and evaluates whether monotonic constraints improve the interpretability and robustness of tree-based models. Using 16,417 firm-year observations from 2014–2024, we estimate LightGBM, XGBoost, and CatBoost with and without monotonic constraints, assess accuracy, precision, recall, F1, and AUC, and conduct rigorous validation using ROC–AUC and precision–recall curves, SHAP explanations, the Kolmogorov–Smirnov and McNemar tests, the Wilcoxon signed-rank test, the 5×2 cross-validation paired t-test, and the Friedman–Nemenyi procedure.

First, to evaluate the effectiveness of integrating sentiment, we compare sentiment-augmented models with financial-only baselines and find consistent performance gains across metrics and tests; features capturing news volume, sentiment change, and news shock provide incremental, forward-looking signal beyond accounting data. Second, to test whether monotonic constraints outperform unconstrained variants, we show that constrained specifications dominate in discrimination, stability, and economic coherence; LightGBM with monotonic constraints achieves accuracy of 0.9853, recall of 99.70%, and AUC of 0.9993 and remains superior under all robustness checks. Third, to identify the factors most strongly affecting default probability, SHAP-based rankings and monotone partial-dependence analyses indicate that leverage measures increase default risk, while stronger liquidity and interest coverage reduce it; profitability and cash-flow capacity further lower risk, and the aforementioned sentiment variables materially elevate or dampen predicted probabilities in line with their direction.

Keywords: Corporate default prediction, Tree-based algorithms, Monotonic constraints, Sentiment analysis, Credit risk modeling.

TÓM TẮT

Luận văn tốt nghiệp này xem xét cách các tỷ lệ tài chính và tâm lý dựa trên tin tức cùng nhau hỗ trợ dự báo vỡ nợ của doanh nghiệp tại các công ty niêm yết tại Việt Nam, đồng thời đánh giá liệu các ràng buộc đơn điệu có cải thiện khả năng diễn giải và độ mạnh của các mô hình cây hay không. Sử dụng 16.417 quan sát theo năm công ty từ năm 2014–2024, chúng tôi ước tính LightGBM, XGBoost và CatBoost có và không có ràng buộc đơn điệu, đánh giá độ chính xác, độ chính xác, độ thu hồi, F1 và AUC, đồng thời tiến hành kiểm định nghiêm ngặt bằng cách sử dụng các đường cong ROC–AUC và độ chính xác–thu hồi, giải thích SHAP, kiểm định Kolmogorov–Smirnov và McNemar, kiểm định thứ hạng có dấu Wilcoxon, kiểm định t ghép đôi xác thực chéo 5×2 và quy trình Friedman–Nemenyi.

Đầu tiên, để đánh giá hiệu quả của việc tích hợp tâm lý, chúng tôi so sánh các mô hình tăng cường tâm lý với các đường cơ sở chỉ liên quan đến tài chính và nhận thấy hiệu suất tăng trưởng nhất quán qua các chỉ số và kiểm định; các tính năng nắm bắt khối lượng tin tức, thay đổi tâm lý và cú sốc tin tức cung cấp tín hiệu gia tăng, hướng tới tương lai vượt ra ngoài dữ liệu kế toán. Thứ hai, để kiểm tra xem các ràng buộc đơn điệu có vượt trội hơn các biến thể không bị ràng buộc hay không, chúng tôi chỉ ra rằng các thông số ràng buộc chiếm ưu thế về khả năng phân biệt, tính ổn định và tính nhất quán kinh tế; LightGBM với các ràng buộc đơn điệu đạt độ chính xác 0,9853, độ thu hồi 99,70% và AUC 0,9993, và vẫn vượt trội hơn trong tất cả các kiểm tra độ mạnh. Thứ ba, để xác định các yếu tố ảnh hưởng mạnh nhất đến xác suất vỡ nợ, xếp hạng dựa trên SHAP và phân tích phụ thuộc một phần đơn điệu cho thấy các biện pháp đòn bẩy làm tăng rủi ro vỡ nợ, trong khi thanh khoản và khả năng bao phủ lãi suất mạnh hơn làm giảm rủi ro; lợi nhuận và khả năng dòng tiền tiếp tục làm giảm rủi ro, và các biến số tâm lý đã đề cập ở trên làm tăng hoặc giảm đáng kể các xác suất dự đoán theo hướng của chúng.

Từ khóa: Dự báo vỡ nợ doanh nghiệp, Thuật toán dựa trên cây, Ràng buộc đơn điệu, Phân tích cảm xúc, Mô hình hóa rủi ro tín dụng.

COMMITMENT

I hereby declare that the graduation thesis entitled “Sentiment-Enhanced Machine Learning for Corporate Default Prediction: A Tree-Based Approach with Monotonic Constraints” is the result of my own independent research. All data, analyses, and findings presented in this thesis are truthful and have not been previously published or conducted by any other individual or organization, except for properly cited sources as referenced throughout the thesis.

Ho Chi Minh City, September, 30th, 2025

Author

Ngo Hoang Khanh Duy

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Lastly, I acknowledge that due to my limited knowledge and experience, there may still be shortcomings in this thesis. I sincerely welcome any comments and suggestions from esteemed lecturers and readers to further improve the quality of this research.

Best Regard,

Ngo Hoang Khanh Duy

TABLE OF CONTENTS

ABSTRACT	i
TÓM TẮT	ii
COMMITMENT	iii
ACKNOWLEDGMENTS.....	iv
LIST OF ABBREVIATIONS.....	viii
LIST OF TABLES.....	ix
LIST OF FIGURES.....	x
CHAPTER 1: INTRODUCTION	1
1.1. Research rationale	1
1.2. Research objectives	3
1.2.1. General objective	3
1.2.2. Specific objectives	4
1.3. Research questions	4
1.4. Research subjects and scope.....	4
1.5. Research methods and data	5
1.5.1. Research methods	5
1.5.2. Research data	5
1.6. Contribution of the research	5
1.7. Structure of the research.....	7
CONCLUSION OF CHAPTER 1	9
CHAPTER 2: THEORETICAL FRAMEWORK AND LITERATURE REVIEW..	10
2.1. Theoretical framework	10
2.1.1. Concept of Corporate Default Risk	10
2.1.2. Measurement of Corporate Default Risk.....	11
2.1.3. Concept of sentiment analysis in credit risk modeling.....	13
2.1.4. Probability of Default Models	14
2.1.5. Monotonic Constraints and Corporate Consistency	16
2.2. Literature review	18

2.2.1. Domestic studies	18
2.2.2. International studies	20
2.3. Novelty of the research	27
CONCLUSION OF CHAPTER 2	29
CHAPTER 3: RESEARCH METHODOLOGY	30
3.1. Research Design	30
3.1.1. Sentiment Data Processing	30
3.1.2. Financial Data Processing	33
3.1.3. Corporate Default Prediction Pipeline	35
3.2. Data collection and processing	40
3.2.1. Financial Data	40
3.2.2. Non-financial Data	43
3.3. Selection of input variables in the default prediction model	44
3.4. Models for predict the probability of default	51
3.4.1. LightGBM	51
3.4.2. XGBoost	53
3.4.3. CatBoost	55
3.5. Methods for selecting a model to predict default probability	57
3.5.1. Testing and evaluating the model on out-of-sample data	57
3.5.2. Criteria to measure the model's ability to predict default risk	58
3.6. Nonparametric and Paired Statistical Tests	61
3.6.1. Kolmogorov–Smirnov (two-sample, distributional separation)	61
3.6.2. McNemar Test (paired error proportions on the same cases)	61
3.6.3. Wilcoxon Signed-Rank (paired metric differences)	62
3.6.4. 5×2 CV Paired t-Test (two algorithms, variance-aware)	62
3.6.5. Friedman Ranks (omnibus, kkk models over NNN matched datasets)	62
3.6.6. Post-hoc Nemenyi (all-pairs after Friedman)	63
CONCLUSION OF CHAPTER 3	64
CHAPTER 4: EMPIRICAL RESULTS AND DISCUSSION	65

4.1. Descriptive statistics of variables	65
4.2. Correlation matrix	69
4.3. Regression results of parametric models.....	70
4.3.1. XGBoost	70
4.3.2. LightGBM.....	78
4.4. Comparison results on the predictive ability of the models	84
4.4.1. Estimation results of the models.....	84
4.4.2. Results of analyzing the importance of variables	90
4.5. Effect of monotonicity constraints	93
4.6. Model Comparison through Statistical Testing	96
CONCLUSION OF CHAPTER 4.....	102
CHAPTER 5: CONCLUSION AND IMPLICATIONS	103
5.1. Conclusion	103
5.2. Implications	104
5.3. Limitations and directions for future research	105
5.3.1. Limitations of the research	105
5.3.2. Directions for future research	106
CONCLUSION OF CHAPTER 5.....	107
REFERENCES.....	
APPENDIX	

LIST OF ABBREVIATIONS

Abbreviation	Abbreviated words
AUC	Area Under the Curve
CDS	Credit default swaps
EBIT	Earnings Before Interest and Taxes
EBITDA	Earnings Before Interest, Taxes, Depreciation, and Amortization
EFB	Exclusive Feature Bundling
FPR	False Positive Rate
GOSS	Gradient-based One Side Sampling
HOSE	Ho Chi Minh Stock Exchange
HNX	Hanoi Stock Exchange
HTML	Hyper Text Markup Language
LR	Logistic Regression
ML	Machine Learning
NER	Named Entity Recognition
PD	Probability of default
P2P	Peer-to-Peer
RF	Random Forest
ROA	Return on Assets
ROC	Receiver Operating Characteristic
ROE	Return on Equity
SBV	State Bank of Vietnam
SHAP	SHapley Additive exPlanations
SMEs	Small and medium-sized enterprises
SVM	Support Vector Machine
TPE	Tree-structured Parzen Estimator
TPR	True Positive Rate
UPCOM	Unlisted Public Company Market

LIST OF TABLES

Table 2.1: Summary of previous studies	23
Table 3.1: Monotonic Constraints and Relationships	37
Table 3.2: Summary of the Number of Enterprises – Business Sectors	41
Table 3.3: Separate default and non-default enterprises.....	42
Table 3.4: Summary of variables	48
Table 3.5: Confusion Matrix	58
Table 3.6: Metrics for each class in the Confusion Matrix.....	59
Table 3.7: Groups of indicators to evaluate the reliability of a model	59
Table 4.1: Descriptive statistics of variables	65
Table 4.2: Default prediction results of models on out-of-sample datasets	85
Table 4.3: Kolmogorov–Smirnov and McNemar Tests for Sentiment Comparisons	96
Table 4.4: Wilcoxon Signed-Rank Test and 5x2 CV Paired t-test.....	98
Table 4.5: Friedman Ranks and Post-hoc Nemenyi Test.....	100

LIST OF FIGURES

Figure 3.1: Flowchart for Named Entity Recognition (NER) and Keyword Matching	31
Figure 3.2: Sentiment Analysis Process with Underthesea.....	32
Figure 3.3: Models Training Process for Predicting Corporate Default.....	40
Figure 3.4: Data source description for the news classification model	43
Figure 3.5: Decision Tree Algorithm Based on Frequency Histogram.....	52
Figure 3.6: Level-wise and Leaf-wise Tree Construction Methods	53
Figure 3.7: Schematic diagram of the XGBoost regression tree model	54
Figure 3.8: The structure of the CatBoost algorithm	56
Figure 3.9: Simulation of AUC.....	61
Figure 4.1: Correlation matrix.....	70
Figure 4.2: Regression result of XGBoost – No Constraints	70
Figure 4.3: Regression result of XGBoost – Constraints.....	74
Figure 4.4: Regression result of LightGBM – No Constraints.....	78
Figure 4.5: Regression result of LightGBM – Constraints	81
Figure 4.6: Prediction results of models on the Confusion matrix	87
Figure 4.7: Prediction results of models on ROC Curves	88
Figure 4.8: Prediction results of models on Precision-Recall Curves.....	89
Figure 4.9: SHapley Additive exPlanations (SHAP)	90
Figure 4.10: Partial dependence plots	93

CHAPTER 1: INTRODUCTION

1.1. Research rationale

Accurate corporate default prediction underpins financial stability, investor confidence, and efficient capital allocation, especially in markets where informational frictions are material. According to Sigrist and Leuenberger (2023), default risk is inherently dynamic and correlated across firms through latent macro-frailty, so models that account for multi-period risk deliver more decision-relevant signals for portfolio management and regulation. In practice, recent evidence shows that state-of-the-art machine learning can materially improve the discrimination and calibration of corporate default forecasts relative to legacy statistical tools, strengthening early-warning capabilities for lenders, investors, and supervisors (Moscatelli et al., 2020).

Traditional models such as logistic regression, discriminant analysis, and Z-score variants remain valued for transparency, yet their linear structures, narrow covariate sets, and reliance on structured accounting ratios constrain what they can discover. Moscatelli et al. (2020) argue that these models struggle with complex non-linearities and interaction effects that characterize distress propagation through business cycles. At the same time, a growing body of empirical finance shows that forward-looking information embedded in narratives and market commentary contains incremental credit-risk content that conventional financial statements do not capture. Smales (2016) documents that news sentiment co-moves with bank credit risk, indicating a channel through which textual signals can lead market-based default perceptions.

To address these limitations, recent research integrates sentiment features with tree-based learning while imposing economically motivated constraints to preserve interpretability. Bie et al. (2025) show that news-based variables extracted with contemporary language models significantly improve bankruptcy prediction over financial ratios alone and that full-text processing outperforms headline signals. Ardekani et al. (2024) argue that domain-tuned large language models can

deliver higher-fidelity financial sentiment, improving the signal-to-noise ratio of text features. Within the modeling layer, monotonic constraints embed financial logic by ensuring that risk moves in theoretically consistent directions with respect to key predictors; Chen and Li (2014) demonstrate that monotone constraints enhance both plausibility and performance in credit rating, while Feelders and Pardoel (2003) and González et al. (2015) provide foundational methods for monotone trees and forests that maintain order-preserving relationships without sacrificing predictive strength.

However, incorporating unstructured text into default prediction raises nontrivial practical and econometric challenges. First, data acquisition and curation are costly, and coverage can be uneven across issuers and languages, heightening the risk of selection biases; Ardekani et al. (2024) discuss domain adaptation and multilingual alignment as persistent hurdles for sentiment engines. Second, temporal alignment is delicate because default events are rare and reported with lags, whereas news arrives at high frequency; Bie et al. (2025) explicitly address horizon design and show that daily news must be aggregated carefully to avoid look-ahead bias while preserving timeliness. Third, label scarcity and class imbalance complicate estimation and evaluation, with post-hoc explainability tools sometimes misleading inference if used incautiously; Carmona et al. (2022) argue for transparent, feature-attribution workflows in distress modeling to guard against spurious correlations. Fourth, measurement error in sentiment extraction and shifts in lexical context across regimes can attenuate signal quality, a concern also noted in earlier evidence linking news tone with credit risk premia (Smales, 2016).

In this study, we develop a tree-based framework that integrates high-frequency sentiment with traditional financials and market-based indicators while imposing monotonic constraints aligned with financial theory. The framework targets publicly listed Vietnamese firms on HOSE, HNX, and UPCoM, where disclosure heterogeneity and sentiment-sensitive trading can amplify informational asymmetries. According to Sigrist and Leuenberger (2023), modeling forward

default probabilities across horizons enhances portfolio and policy relevance; we therefore evaluate one-quarter to one-year horizons, compare unconstrained and constrained gradient-boosted trees, and quantify the incremental value of sentiment. Consistent with Chen and Li (2014), monotonicity is encoded for predictors whose directional effects are theoretically settled, sustaining economic coherence without sacrificing accuracy. We expect three outcomes: first, sentiment-enhanced specifications will outperform ratio-only baselines in discrimination and early-warning timeliness; second, monotonic constraints will improve robustness and trustworthiness by eliminating counterintuitive partial effects; third, global and local explanations from constrained trees will identify the most influential factors shaping default probabilities under Vietnam’s evolving market conditions.

This thesis, titled “**Sentiment-Enhanced Machine Learning for Corporate Default Prediction: A Tree-Based Approach with Monotonic Constraints**”, aims to develop a predictive framework tailored to Vietnamese listed companies. By combining sentiment-based variables with traditional financial indicators in tree-based models and applying monotonic constraints, the study seeks to deliver a more accurate, interpretable, and economically consistent default prediction tool. This framework is intended to enhance early warning systems and support financial institutions, investors, and policymakers in building a more resilient and forward-looking default risk assessment environment.

1.2. Research objectives

1.2.1. General objective

The primary objective of this study is to investigate the effectiveness of integrating sentiment information into tree-based machine learning models for corporate default prediction in the Vietnamese market. Specifically, the research aims to compare the predictive performance and interpretability of constrained and unconstrained models, while ensuring consistency with established financial logic. In doing so, the study also seeks to identify and assess the most influential financial and sentiment-driven factors shaping corporate default risk, thereby contributing to

more robust and economically coherent default risk assessment frameworks in Vietnam's capital markets.

1.2.2. Specific objectives

To evaluate the effectiveness of incorporating sentiment data into machine learning models for enhancing corporate default prediction accuracy.

To examine whether models with monotonic constraints outperform unconstrained models in terms of predictive performance, corporate consistency, and interpretability (Suárez et al., 2024).

To identify and assess the most influential financial and sentiment-related factors shaping corporate default risk within the developed models.

1.3. Research questions

To what extent does the integration of sentiment information improve the predictive accuracy of tree-based machine learning models for corporate default risk in the Vietnamese market?

Do models with monotonic constraints provide superior predictive performance, interpretability, and corporate consistency compared to that of unconstrained?

Which financial and sentiment-related factors exert the most significant influence on the probability of corporate default within the developed prediction framework?

1.4. Research subjects and scope

Research subjects: The central focus of this thesis is the development and evaluation of sentiment-enhanced tree-based machine learning models for predicting corporate default risk, with emphasis on how the integration of sentiment analysis and monotonic constraints can improve the accuracy, interpretability, and corporate consistency of default prediction for Vietnamese listed companies.

Scope of research:

Temporal scope: The analysis covers the period from 2014 to 2024, a decade that provides a sufficiently updated and comprehensive perspective on

default risk. The chosen timeframe not only reflects the most recent ten years of market development but also encompasses major economic fluctuations and structural changes affecting Vietnamese listed firms. By spanning multiple phases of growth and volatility, this horizon allows for a rigorous evaluation of how financial and sentiment-based indicators interact in shaping corporate default risk.

Spatial scope: The sample comprises publicly listed companies on Vietnam's stock exchanges, including the Ho Chi Minh Stock Exchange (HOSE), the Hanoi Stock Exchange (HNX), and the Unlisted Public Company Market (UPCoM).

1.5. Research methods and data

1.5.1. Research methods

Qualitative Method: A comprehensive literature review and theoretical synthesis are conducted to establish the conceptual framework, define hypotheses, and determine economically consistent monotonic relationships between predictor variables and default probability.

Quantitative Method: Sentiment-enhanced tree-based machine learning models (XGBoost, LightGBM, CatBoost) are developed with and without monotonic constraints. Model performance is evaluated using cross-validation and metrics such as F1-score, ROC-AUC, and Precision-Recall AUC, with SHAP analysis applied to identify the most influential financial and sentiment-based variables.

1.5.2. Research data

This study uses secondary data comprising financial indicators from audited reports of Vietnamese listed companies on HOSE, HNX, and UPCoM obtained via the VNStock platform, and sentiment scores extracted from financial news and corporate announcements sourced from Vietstock, processed using natural language processing techniques.

1.6. Contribution of the research

Theoretical contributions

This study contributes to the academic discourse on corporate default prediction and machine learning applications in risk assessment by situating the analysis within the context of Vietnamese listed firms, an emerging market where both financial fundamentals and market sentiment shape default dynamics. It advances theory in two key directions. First, it demonstrates how sentiment signals derived from financial news and market communication can complement traditional financial indicators, thereby capturing the behavioral and informational dimensions of default risk that conventional accounting-based models often overlook. Second, it highlights the importance of monotonic constraints in tree-based models as a mechanism for embedding financial logic, ensuring that predictive outputs remain both economically consistent and interpretable. By systematically comparing constrained and unconstrained models, the study elucidates how sentiment features interact with financial ratios under different modeling assumptions, offering novel insights into the integration of unstructured data and constraint-based machine learning. In doing so, it enriches theoretical understanding of default risk modeling in emerging markets, where information asymmetry and volatility make sentiment particularly salient.

Practical contributions

This study provides empirical insights for financial institutions, investors, and market regulators in Vietnam to enhance default risk assessment practices through the integration of sentiment analysis and monotonic constraints in machine learning models. It offers evidence-based guidance on developing predictive tools that are both accurate and economically consistent, enabling earlier detection of potential defaults among listed companies. The findings support the design of more transparent, interpretable, and market-responsive risk assessment frameworks, helping stakeholders strengthen early warning systems and improve decision-making in Vietnam's capital markets.

1.7. Structure of the research

This thesis is structured into five main chapters, each of which contributes systematically to the research objectives:

CHAPTER 1: INTRODUCTION

This chapter presents the research background and outlines the research objectives, questions, scope, methods, and expected contributions. It also provides an overview of the thesis layout.

CHAPTER 2: THEORETICAL FRAMEWORK AND LITERATURE REVIEW

This chapter introduces the key theoretical foundations underpinning the research. It also reviews both domestic and international empirical studies related to board characteristics and bank stability, thereby identifying research gaps and supporting the development of the proposed model and hypotheses.

CHAPTER 3: RESEARCH METHODOLOGY

This chapter outlines the research design, the construction of the empirical model, the definitions and measurement of variables, the research hypotheses, the data collection process, and the econometric techniques used for analysis.

CHAPTER 4: EMPIRICAL RESULTS AND DISCUSSION

This chapter presents descriptive statistics, model evaluation results, and comparative analysis between constrained and unconstrained models. It interprets the findings, examines the influence of financial and sentiment-based variables on default risk, and discusses the implications of integrating monotonic constraints, with a focus on the Vietnamese capital market context.

CHAPTER 5: CONCLUSIONS AND POLICY IMPLICATIONS

The final chapter summarizes the key findings, discusses the research limitations, and provides practical recommendations for financial institutions, investors, and market regulators in Vietnam. It also suggests directions for future research to advance the application of sentiment analysis and monotonic constraints in prediction default modeling.

CONCLUSION OF CHAPTER 1

This chapter has laid the foundation for the thesis by presenting the background, rationale, and significance of developing sentiment-enhanced machine learning models with monotonic constraints for predicting corporate default risk in the context of Vietnamese listed companies. It has defined the research problem, articulated the general and specific objectives, and outlined the key research questions to be addressed. The scope of the study has been specified, including the spatial focus, data sources from VNStock and Vietstock, and the integration of both qualitative and quantitative methods. The chapter has also highlighted the expected theoretical contributions in advancing the literature on prediction default modeling and the practical implications for enhancing risk assessment frameworks in Vietnam's capital markets. Finally, an overview of the thesis structure has been provided. The next chapter introduces the theoretical framework underpinning the study and reviews relevant domestic and international literature on corporate default prediction, sentiment analysis, and monotonic constraints in machine learning.

CHAPTER 2: THEORETICAL FRAMEWORK AND LITERATURE REVIEW

2.1. Theoretical framework

2.1.1. *Concept of Corporate Default Risk*

Corporate default risk is best understood as the likelihood that a firm fails to meet its debt obligations on time, reflecting deterioration in solvency and liquidity. According to Coenen et al. (2022), the probability of default is a forward-looking measure that integrates accounting solvency, cash flow sufficiency, and market perceptions under real-world conditions. Heynderickx et al. (2016) emphasize the distinction between risk-neutral and actual probabilities since pricing-based measures embed premia while real-world probabilities reflect the frequency of failure. Together, these views establish default risk as a stochastic event shaped by firm fundamentals, financing structures, and the macroeconomic environment.

Scholarly research frames default risk as multi-dimensional, incorporating legal, informational, and market-implied perspectives. Yang et al. (2020) argue that credit default swap spreads efficiently aggregate dispersed market information, serving as sensitive barometers when accounting data are outdated. Safiullah et al. (2022) further highlight that model-based measures such as distance-to-default link legal default events to balance sheet dynamics and asset volatility. This view conceptualizes default risk as a continuum of vulnerability observable across accounting, market, and structural indicators.

Default risk is also conditioned by macro-financial uncertainty and information frictions that influence financing costs and resilience. Nguyen et al. (2022) show that uncertainty reduces cash holdings and investment efficiency in emerging markets, heightening vulnerability even when profitability appears intact. Yang et al. (2020) demonstrate that negative news sentiment widens credit spreads, amplifying market-implied default assessments under asymmetric information. These findings suggest default risk evolves with shifting expectations of earnings, refinancing, and shock absorption capacity, not merely with static balance sheet ratios.

At the firm level, the probability of default reflects the interaction of leverage, asset risk, and cash flow sufficiency. Coenen et al. (2022) note that high leverage and volatile earnings raise default risk as thin equity buffers transmit shocks into payment failures. Safiullah et al. (2022) find that innovation, particularly in green technologies, lowers default risk by enhancing efficiency and capital access. Mehmood and De Luca (2023) provide evidence that financial indicators retain explanatory power in private firms, underscoring the relevance of internal performance and financing choices for credit viability.

Finally, recent studies highlight the need to integrate multiple data sources for a more complete representation of default risk. Alvi et al. (2024) argue that combining accounting ratios, market signals, and narrative-based sentiment produces a more accurate construct, provided interpretability is preserved. This approach acknowledges that default is a legal state but emphasizes its economic antecedents as continuous and time-varying, shaped by flows of financial and informational signals.

2.1.2. Measurement of Corporate Default Risk

Measuring corporate default risk requires translating a firm's evolving financial condition into quantifiable indicators of insolvency. Altman et al. (2017) argue that robust measurement frameworks must integrate firm-specific ratios, market indicators, and macroeconomic conditions to capture both capacity and willingness to meet obligations. Coenen et al. (2022) emphasizes that models should estimate probabilities of default (PD) over specified horizons rather than only predict binary failure events, thereby enhancing proactive mitigation. In this sense, measurement balances accuracy, timeliness, and interpretability to inform lending, investment, and regulatory strategies.

Accounting-based models remain widely applied due to accessibility and simplicity. The Altman Z-score combines profitability, leverage, liquidity, solvency, and activity ratios to evaluate insolvency risk (Altman et al., 2020), while the Ohlson O-score employs logistic regression with size, leverage, and performance

variables. Mehmood and De Luca (2023) highlight their relevance in emerging markets such as Vietnam, where liquidity constraints reduce the reliability of market indicators. Yet Nguyen et al. (2022) caution that earnings manipulation and accounting flexibility can obscure gradual financial deterioration, limiting their predictive strength.

Market-based measures introduce a forward-looking dimension. Heynderickx et al. (2016) note that Merton's distance-to-default, grounded in option pricing, interprets equity as a call option on firm assets with default triggered when asset value falls below debt. Yang et al. (2020) demonstrate that credit spreads, particularly from credit default swaps, integrate dispersed information such as sentiment and macroeconomic expectations into risk assessments. However, Vietnam's limited derivative markets and thin trading volumes restrict their practical use, making hybrid approaches that combine accounting and market data more effective.

Machine learning methods expand these approaches by capturing complex interactions. Safiullah et al. (2022) show that boosting algorithms like XGBoost and LightGBM outperform linear models in default prediction. Alvi et al. (2024) argue that integrating sentiment data from financial news further strengthens predictive accuracy, particularly at short horizons. In Vietnam, where firms are classified at risk of default if they record three consecutive years of losses or face severe capital erosion in bankruptcy proceedings, machine learning models can be calibrated to detect such patterns early and align with regulatory thresholds.

Operational definitions of default also shape measurement in emerging markets. In Vietnam, creditors and courts often identify default risk through cumulative losses over three or more years, persistent negative equity, or loan covenant violations leading to restructuring. Nguyen et al. (2022b) suggest that embedding these jurisdiction-specific criteria into model calibration improves both local relevance and international comparability. Integrating insolvency benchmarks

with financial ratios and sentiment-augmented machine learning thus produces PD estimates that are both empirically rigorous and context-sensitive.

In summary, measurement of corporate default risk synthesizes accounting-based ratios, market-implied indicators, and advanced statistical learning. The most effective approach depends on data quality, market maturity, and regulatory definitions. In Vietnam, combining ratio analysis with sentiment-enhanced machine learning and the three-year loss criterion provides a comprehensive framework. As Coenen et al. (2022) concludes, the ultimate objective is not only accurate detection but also early identification that allows creditors and policymakers to intervene before insolvency becomes irreversible.

2.1.3. Concept of sentiment analysis in credit risk modeling

Sentiment analysis in credit risk modeling extends evaluation beyond financial ratios by systematically extracting meaning from texts. Roeder et al. (2022) note that unstructured sources improve timeliness and context in default assessment. Nguyen and Huynh (2022) argue that disclosure-based sentiment captures managerial tone and forward-looking cues absent in ratios. Yang et al. (2020) further show that firm-specific news sentiment translates into credit default swap (CDS) spreads, demonstrating that textual tone contains tradeable signals of perceived creditworthiness.

Multiple textual streams operationalize sentiment with complementary insights. Hesse and Loy (2025) find that sentence-level cues in Management Discussion and Analysis sections predict bankruptcy when modeled with advanced language tools. Chen et al. (2023) show that integrating annual report tone with financial variables raises predictive accuracy. Y. Chen et al. (2023) highlight that earnings call transcripts and vocal affect embed credit-relevant cues. News sentiment shapes external market views, with Yang et al. (2020) linking negative tone to wider CDS spreads and Wang et al. (2025) showing pessimistic tone increases offering yields. Yao et al. (2023) add that time-weighted social media sentiment enhances listed-firm risk prediction.

Methodologically, sentiment construction spans transparent dictionaries, supervised learning, and deep embeddings. Nguyen and Huynh (2022) argue that finance-specific lexicons yield interpretable tone consistent with disclosure norms. Chen et al. (2023) show that text features in supervised models outperform financial-only baselines. Liu and Jia (2025) contend that hybrid deep learning extracts richer semantics from annual reports. Roeder et al. (2022) emphasize the need for validation frameworks that prevent overfitting when unstructured features are introduced.

The rationale for sentiment integration lies in its strong link to credit outcomes. Yang et al. (2020) show negative news tone coincides with higher CDS spreads, while Wang et al. (2025) find that adverse tone elevates bond yield spreads. Chen et al. (2023) demonstrate that disclosure-based sentiment improves bankruptcy classification, and Hesse and Loy (2025) argue that sentence-level framing uncovers subtle early-warning signals. Wu et al. (2025) suggest that controlled large language model applications can further extract credit signals from heterogeneous texts.

Designing sentiment features requires econometric discipline. Roeder et al. (2022) stress publication-time alignment, out-of-sample testing, and stability across cycles. Alvi et al. (2024) argue for hybrid models combining ratios, markets, and sentiment under strict interpretability safeguards. Liu and Jia (2025) show that deep models can be made transparent with attribution and ablation, allowing links between narrative cues and predicted default risk. Collectively, these findings position sentiment as a validated complement that enhances the responsiveness of credit risk models.

2.1.4. Probability of Default Models

Tree-based machine learning for probability of default has risen from a long tradition of statistical modeling in credit risk. According to Louzada et al. (2016), logistic regression and related linear classifiers long dominated because of parsimony and clear decision logic, yet performance plateaus emerged as data grew

richer and more nonlinear. Brown and Mues (2012) argue that even within classical settings, alternative learners began to close the gap as researchers confronted class imbalance and complex interactions that linear specifications struggled to capture. Akkoç (2012) adds that hybrid and nonlinear methods can extract incremental signal when financial ratios interact in nonadditive ways, foreshadowing the shift toward machine learning.

The first wave of credit-risk machine learning emphasized support vector machines, neural networks, and decision trees as drop-in replacements for logit while scrutinizing sampling bias and imbalance. Brown and Mues (2012) argue that algorithm choice is sensitive to imbalance and population drift, which complicates out-of-sample validity for purely linear scorecards. According to Crone and Finlay (2012), careful instance sampling and rebalancing materially affect comparative accuracy across logistic regression, discriminant analysis, decision trees, and neural networks, underscoring that modeling choices and data engineering co-determine measured gains.

Modern ensemble methods then solidified the advantage of tree-based learners. Lessmann et al. (2015) argue that gradient boosting and related ensembles systematically outperform traditional classifiers in large credit-scoring benchmarks by capturing nonlinearities and higher-order interactions without prespecifying functional forms. Moscatelli et al. (2020) further show that machine learning can improve corporate default forecasting relative to standard statistical baselines when tuned and validated under realistic banking conditions. According to Y. Liu et al. (2022), boosting frameworks deliver higher predictive accuracy and faster screening in applied default prediction tasks, strengthening the case for tree-based PD modeling in practice.

Within this family, recent applications highlight LightGBM, XGBoost, and CatBoost as scalable defaults for PD estimation. Zhu et al. (2023) argue that LightGBM and XGBoost consistently surpass logistic regression and single trees on loan-default tasks while remaining compatible with post-hoc explainability.

According to Y. Liu et al. (2022), XGBoost and related boosting algorithms achieve rapid, accurate, and cost-efficient default prediction in production settings. Extending the toolkit, Xia et al. (2025a) show that CatBoost can be integrated with domain-specific textual features to enhance imbalanced credit scoring while preserving interpretability through SHAP-based decompositions. Together these results trace a coherent progression from linear scorecards to boosted trees as the workhorse for PD.

A parallel literature embeds economic logic via monotonic constraints to improve credibility and governance. Chen and Li (2014) argue that imposing monotonicity in credit models aligns estimated effects with expert priors and reduces counterintuitive patterns that erode trust. According to Pei et al. (2016), decision trees can be trained under monotonic restrictions to respect ordinal relations among risk drivers, while González et al. (2015) extend this principle to random forest ensembles. Bastos and Matos (2022) argues that explainability and economically coherent structure are central to model risk management for credit loss models, which motivates constraint-aware learning in regulated environments.

2.1.5. Monotonic Constraints and Corporate Consistency

In probability-of-default (PD) modeling, monotonic constraints encode economically justified directionality between risk drivers and predicted default probabilities, ensuring that the model’s response aligns with theory and institutional judgment. According to Pei et al. (2016), imposing monotonicity on decision trees restricts the hypothesis space so that increases in specified predictors produce non-decreasing or non-increasing effects on the output, thereby eliminating counterintuitive patterns that often arise in flexible learners trained on noisy financial data. Israeli et al. (2019) argues that constraint-aware boosting provides a principled mechanism for integrating prior knowledge into tree ensembles, improving face validity without excessively sacrificing predictive power. These results position monotonic constraints as an explicit bridge between statistical learning and economic logic in credit risk estimation.

Within tree-based PD models, corporate consistency concerns arise because unconstrained splits can yield locally optimal but globally implausible shapes, for example predicting lower PD at higher leverage for some covariate configurations. Pei et al. (2016) argues that monotone trees address this by rejecting splits that would violate the imposed order, thus preserving a globally consistent response surface across all subtrees. In gradient-boosted settings, Israeli et al. (2019) shows that constraint learning can be embedded directly into the boosting procedure so that each successive tree respects directional requirements, yielding ensembles that remain faithful to economic priors while capturing nonlinear interactions among covariates. According to Doumpos et al. (2023), aligning model structure with domain knowledge is integral to robust banking analytics, as it supports stability across samples and strengthens practitioner trust in model outputs.

Applied to modern boosting frameworks, monotonicity serves two complementary goals: preserving theoretical coherence and enhancing deployability under governance standards. Tang et al. (2023) argue that LightGBM’s gradient-boosted trees are highly effective for corporate credit risk because they detect complex nonlinearities in ratios and cash-flow indicators; incorporating monotonic constraints on leverage or interest-coverage features aligns these gains with expected PD directionality. Bai et al. (2022) show that boosted tree methods also perform strongly in credit survival settings, where economically sensible shapes in hazard responses are essential for rating transitions and early-warning systems. According to Chang et al. (2024), gradient boosting families remain among the leading approaches for credit worthiness assessment, and their acceptance improves when predictive patterns are consistent with financial intuition.

From a modeling-risk perspective, monotonic constraints mitigate well-known vulnerabilities of flexible learners. Israeli et al. (2019) argues that constraints reduce variance and guard against spurious inversions driven by sampling quirks, while retaining much of the discriminatory strength of boosting. Pei et al. (2016) emphasizes that enforcing monotonicity is a global property of the learned tree,

which helps avoid local “shape flips” across adjacent partitions that would otherwise complicate explanation to credit officers and challenge back-testing stability. In credit rating contexts closely related to PD, Chen and Li (2014) show that explicit monotone restrictions on explanatory variables can improve both accuracy and interpretability, suggesting that shape-constrained learning is a general path to economically coherent credit models.

Crucially, economic consistency extends beyond aesthetics to decision usefulness. According to Doumpos et al. (2023), banking models must balance accuracy with transparency to support capital allocation and oversight; monotonicity helps ensure that managerial actions map predictably to PD movements, such as reductions in PD when liquidity buffers increase. Israeli et al. (2019) argues that constraint-aware boosting enables this predictability while still exploiting high-order interactions among financial ratios, market indicators, and cash-flow measures. Pei et al. (2016) adds that, by embedding monotone structure at the split level, tree learners can simultaneously satisfy domain axioms and deliver competitive classification performance, producing PD surfaces that are stable, interpretable, and aligned with credit economics.

2.2. Literature review

2.2.1. *Domestic studies*

Research on corporate default prediction in Vietnam has developed along three interconnected strands, namely the advancement of accounting- and market-based distress models, the construction of sentiment measures tailored to the Vietnamese information environment, and the application of tree-based machine learning algorithms for probability of default estimation. According to Pham Vo Ninh et al. (2018), one of the earliest comprehensive frameworks for default prediction in Vietnam integrated accounting, market, and macroeconomic indicators to identify financially distressed listed firms, providing a foundational benchmark for subsequent research. Building on this foundation, Truong and Le (2022) argue that ensemble tree classifiers surpass logistic regression in bankruptcy prediction

using Vietnamese firm-level data, with boosting and bagging methods demonstrating superior accuracy and balanced error rates. These studies collectively highlight Vietnam as a promising environment where richer data sources and non-linear learners can be leveraged to improve the precision of probability-of-default estimation.

Alongside the evolution of modeling techniques, a stream of literature has emerged around sentiment construction adapted to Vietnamese data contexts, offering textual variables that could complement traditional financial predictors in default models. Ha (2025) develops a Vietnam-specific investor sentiment index and demonstrates that it exhibits both contemporaneous and lead–lag connectedness with exchange rate dynamics, revealing macro-financial channels through which sentiment impacts the economy. Nguyen et al. (2025) extracts sentiment from over a decade of Vietnamese Facebook data and argues that social-media tone has both short-run and long-run associations with the VN-Index, suggesting its potential as a forward-looking risk indicator. Extending to the corporate debt market, Nguyen et al. (2024) show that investor sentiment, proxied by Google search intensity in real estate and bond topics, influences the growth trajectory of Vietnam’s real-estate corporate bond market, indicating that credit conditions are partly sentiment-driven. Together, these works confirm the feasibility and informational relevance of Vietnamese textual sentiment for credit risk modeling.

Firm-level distress modeling in Vietnam has also embraced more advanced feature selection and specification methods, though integration with sentiment remains uncommon. Vu et al. (2023) applies a LASSO-based methodology to select a parsimonious set of predictors for financial distress among Vietnamese listed firms, reporting improved model stability compared to traditional ratio-based approaches. Phan et al. (2022) emphasize the significant explanatory power of cash-flow-based variables in predicting distress across private listed enterprises in Vietnam, underscoring the enduring importance of liquidity and profitability measures in an emerging-market setting. However, these approaches are

predominantly accounting-focused and do not yet incorporate complementary sentiment signals.

Tree-based machine learning methods have been validated in the Vietnamese credit risk context and represent a natural foundation for sentiment-enhanced default modeling. Nguyen and Ngo (2025) provide a comparative evaluation of boosting algorithms for predicting personal default and conclude that gradient boosting methods consistently outperform baseline statistical models in terms of predictive accuracy and robustness. Although their study is centered on personal credit, the authors contend that the adaptability of boosting algorithms to complex non-linear relationships makes them equally applicable to corporate default prediction in heterogeneous and evolving data environments like Vietnam. This perspective aligns with Truong and Le (2021), who demonstrate the effectiveness of tree-based approaches in bankruptcy prediction for Vietnamese firms, and with Duc et al. (2024) who find that macroeconomic uncertainty significantly magnifies corporate default risk in emerging markets, thereby requiring modeling architectures capable of capturing non-linear macro-financial interactions.

2.2.2. International studies

A rapidly expanding international literature shows that integrating unstructured textual signals into probability-of-default (PD) modeling improves timeliness and discriminatory power relative to ratio-only baselines. According to Alvi et al. (2024), recent default-prediction research converges on hybrid architectures that fuse financial ratios, market variables, and text-derived sentiment, provided models preserve economic interpretability. Roeder et al. (2022) argues that unstructured data such as analyst narratives and issuer disclosures carry incremental information for credit risk, motivating model designs that admit text features alongside traditional covariates. Together, these reviews establish the conceptual foundation for sentiment-enhanced PD modeling and frame the need for algorithms that can ingest heterogeneous inputs while remaining explainable.

Empirically, sentiment extracts from external news and issuer communications map into credit spreads and default risk, validating their use as PD features. Yang et al. (2020) show that firm-specific negative news sentiment widens CDS spreads, consistent with higher market-implied default probabilities and the information role of public news under asymmetry. Chen et al. (2023) demonstrate that the communicative value and tone in annual reports measurably enhance bankruptcy-prediction accuracy when combined with machine-learning classifiers. Extending to spoken disclosures, Chen et al. (2023) find that emotional cues in earnings-call audio shape market reactions, implying that multimodal sentiment can proxy evolving credit quality. These studies provide the empirical channel through which sentiment augments PD estimation.

On the modeling side, gradient-boosted decision trees have become the workhorse for PD because they capture nonlinearities and high-order interactions in financial data at scale. Lessmann et al. (2015) document the strong performance of modern tree ensembles in credit scoring relative to classical methods. According to Sun et al. (2022), LightGBM paired with imbalance-aware bagging delivers robust gains for enterprise credit evaluation, while Liu et al. (2022) show that XGBoost-based two-stage hybrids improve credit-risk prediction over single-model baselines. Complementary evidence in borrower datasets indicates that boosted trees remain competitive across timing structures and data regimes, reinforcing their suitability as the backbone for sentiment-enhanced PD pipelines.

A central challenge is aligning powerful tree ensembles with economic priors so that PD responds in the expected direction to core risk factors. Monotonic constraints address this by restricting the learned response function to be globally non-decreasing or non-increasing in prespecified predictors. Pei et al. (2016) propose decision trees with monotonicity constraints that prevent local “shape flips,” thereby improving face validity without sacrificing much accuracy. Israeli et al. (2019) embed constraint learning directly into gradient boosting, showing that ensembles can satisfy directional requirements with minimal loss in overall

predictive performance. In PD applications, such constraints operationalize domain knowledge, for example enforcing that higher leverage or lower interest-coverage cannot decrease the estimated default probability.

These strands intersect naturally when sentiment features are introduced into tree-based PD models. Reviews of recent credit-scoring practice emphasize boosted trees as state-of-the-art while calling for designs that remain interpretable and governance-ready; monotonic constraints supply that structure, and sentiment supplies forward-looking information. Markov et al. (2022) argue that contemporary credit-scoring systems increasingly integrate alternative data and advanced learners, but require techniques that guard against economically implausible patterns. In parallel, borrower-level and platform studies confirm the external validity of boosted trees across credit contexts, supporting their role as integrators of structured ratios and text-based sentiment. When CatBoost, LightGBM, or XGBoost are deployed with monotonicity on key financial drivers, the resulting PD surfaces remain consistent with theory while exploiting the incremental signal in sentiment.

Recent applications further illustrate how these ingredients can be combined in practice. Sun et al. (2022) show LightGBM's efficacy under class imbalance, suggesting a natural host for sentiment variables that are timely but noisy. Liu et al. (2022) use XGBoost as a feature-transformer within a hybrid pipeline, a design that can be extended to include sentiment embeddings. Meanwhile, work on fintech credit scoring integrates CatBoost with auxiliary representations, indicating that category-rich boosting frameworks can stably absorb heterogeneous inputs, including text-derived scores. Collectively, the international evidence supports a modeling agenda that fuses sentiment features with tree-based learners under monotonic constraints to deliver PD estimates that are both predictive and economically coherent.

Below is a summary table of literature review:

Table 2.1: Summary of previous studies

Authors	Research data	Research method	Main Findings
<i>Domestic studies</i>			
Ninh et al. (2018)	Vietnamese listed firms, 2003–2016	Integrated accounting, market, and macro indicators; AUC comparisons across model families	Comprehensive Vietnam-specific framework; accounting factors dominate in the combined model for distress prediction.
Truong & Le (2022)	300 Vietnamese firms, 2017–2019	Comparative ML vs. logistic; ensemble trees (e.g., Random Forest)	Intelligent approaches outperform logistic regression; Random Forest yields best accuracy and error balance.
Ha (2025)	Vietnam investor sentiment index and exchange rates, daily data 2017-01-01 to 2023-11-25	TVP-VAR with stochastic volatility; connectedness analysis	Significant contemporaneous and lead–lag connectedness between sentiment and FX dynamics.
H. H. Nguyen et al. (2025)	~773,000 curated Vietnamese Facebook	VECM for long-run dynamics; logistic and quantile	Social-media sentiment predicts returns across horizons; stronger effects in lower return quantiles.

	posts, 2013–2023; VN-Index	regressions; multiple ML regressors	
N. T. Nguyen et al. (2024)	Vietnam real-estate corporate bond market; quarterly data, 54 observations	ARDL; Google Trends–based sentiment index construction	Real-estate sentiment boosts bond-market growth; stock-market sentiment dampens it; long-run negative effect from bond-sentiment on issuance value.
Vu et al. (2023)	492 Vietnamese listed firms, 2012–2021	LASSO-based predictor selection with ML classification	Parsimonious set of ratios improves model stability and distress prediction during COVID-era window. IDEAS/RePEc
Phan et al. (2022)	263 private non-financial listed enterprises (HOSE & HNX), 2010–2020	Quantile regression on cash-flow components	Operating and financing cash flows reduce distress; investing cash flow associates with higher distress. IDEAS/RePEc
Nguyen & Ngo (2025)	7,542 Vietnamese borrowers from banks and finance	Comparative boosting: AdaBoost, XGBoost, LightGBM,	Boosting algorithms consistently outperform baseline statistical models on personal default prediction. Taylor & Francis OnlineResearchGate

	companies, 2014–2022; 12 features	CatBoost	
D. N. Nguyen et al. (2022b)	26 emerging markets (including Vietnam), 1991–2019	Panel estimators; firm default risk via distance-to- default	Higher uncertainty increases default risk, especially in tails; effects moderated by firm size and market development. IDEAS/RePEc
<i>International studies</i>			
Alvi et al. (2024)	Global studies on default- prediction models (2015– 2024)	Systematic literature review	Comprehensive synthesis of DPM techniques—statistical, hybrid, textual models—with insights into future research directions.
Chen et al. (2023)	U.S. firms’ annual reports, 1994–2018	ML (LR, RF, XGBoost, SVM) augmented with communicative value	Incorporating text-based communicative value significantly improves prediction accuracy and reduces Type II error, especially for RF and XGBoost.
Chen et al. (2023)	Earnings-call audio from listed firms	Deep learning emotional cue extraction + market reaction analysis	Managerial emotional tone influences analysts’ behavior and market reactions, offering incremental predictive insight.
Israeli et al. (2019)	Benchmark datasets across	Constraint-based gradient boosting	Monotonic constraints enforce economic coherence in gradient

	domains	trees	boosting without sacrificing predictive performance.
Lessmann et al. (2015)	Eight credit scoring datasets (includes European, PAKDD, UCI)	Benchmark of 41 classifiers	Ensemble tree methods perform best among evaluated models across diverse datasets.
Liu et al. (2022)	U.S. LendingClub loan data, 2007–2016	Two-stage hybrid: XGBoost → graph-based deep NN	Superior credit-risk prediction achieved using hybrid pipeline over single-model baselines.
Pei et al. (2016)	Chinese corporate sector datasets	Decision trees with monotonicity constraints	Constraints enhance interpretability and maintain accuracy by preventing illogical predictor–outcome relationships.
Sun et al. (2022)	Chinese enterprise bond issuers, 2014–2020	Asymmetric bagging + LightGBM	Efficiently addresses imbalance in multi-class credit evaluation, improving minority-class prediction.
Xia et al. (2025)	Fintech lending data with narrative inputs	LLM preprocessing + CatBoost with FocalPoly loss	LLM-enhanced features and focal loss significantly improve model performance under class imbalance.
Yang et al. (2020)	Firm-level news sentiment &	Panel regression with sentiment	Negative firm news increases CDS spreads, especially under high information asymmetry.

	CDS spreads, Jan 2001–Sep 2016		
Zhou et al. (2021)	P2P lending data with two timestamps	Gradient boosting with time-aware scheme	Two-timestamp evaluation better captures temporal dynamics of default risk.

Source: Compiled by the author based on reviewed literature

2.3. Novelty of the research

Although corporate default prediction has been extensively examined in the global credit risk literature, important limitations remain, particularly in integrating sentiment analysis with tree-based machine learning models under monotonic constraints. This study addresses these gaps and introduces novel contributions in several dimensions.

First, the research provides a focused investigation on the Vietnamese corporate sector, a rapidly evolving emerging market where financial reporting transparency, market sentiment, and credit risk management practices are still developing. While many existing studies have applied sentiment-augmented credit risk models in mature economies, empirical applications in Vietnam are scarce, especially those that combine textual sentiment features with structured financial data in predictive modeling.

Second, the study integrates sentiment analysis into probability-of-default estimation using only tree-based algorithms that natively support monotonic constraints is LightGBM, XGBoost, and CatBoost. By enforcing monotonic relationships between key financial risk factors and default probability, the research ensures economic consistency in predictions while capturing the incremental, forward-looking information embedded in sentiment variables. This methodological combination remains largely unexplored in prior credit risk studies, particularly in emerging market contexts.

Third, the research constructs a unique, multi-source dataset that merges firm-level financial statements, macroeconomic indicators, and sentiment measures extracted from Vietnamese financial news and market communications. This dataset covers the period from 2014 to 2024, including the post-pandemic years when shifts in investor sentiment, regulatory interventions, and corporate restructuring activity were most pronounced. Incorporating sentiment derived from local-language sources enhances contextual relevance and reflects nuances in domestic market perception often overlooked by international sentiment dictionaries.

In summary, this study makes a distinctive contribution to the credit risk modeling literature by delivering a sentiment-enhanced, tree-based default prediction framework with monotonic constraints, tailored to the Vietnamese corporate environment. The integration of forward-looking sentiment signals, theory-aligned monotonic restrictions, and a locally sourced dataset offers both methodological innovation and context-specific insight, advancing predictive accuracy and interpretability in corporate default risk assessment.

CONCLUSION OF CHAPTER 2

In Chapter 2, the author presents a comprehensive review of the theoretical foundations and prior empirical studies relevant to sentiment-enhanced corporate default prediction using tree-based machine learning models with monotonic constraints. The chapter discusses key concepts including corporate default risk, methods for its measurement, the role of sentiment analysis in credit risk modeling, probability-of-default estimation using gradient-boosted tree algorithms, and the application of monotonic constraints to ensure economic consistency in predictive outputs.

Both domestic and international studies are examined to illustrate how sentiment-derived variables, when integrated with financial and macroeconomic indicators, can enhance the predictive performance of LightGBM, XGBoost, and CatBoost models. The review also highlights how monotonic constraints have been used to align machine learning predictions with theoretical expectations in credit risk contexts, while noting that such integration remains underexplored in emerging markets.

The author identifies clear gaps in the literature: limited empirical applications of sentiment-enriched PD modeling in the Vietnamese corporate sector, scarce use of monotonic constraints in local credit risk research, and a lack of context-specific datasets that combine structured financial data with local-language sentiment indicators. Addressing these gaps forms the basis for developing the research model and methodological framework in Chapter 3, ensuring both methodological innovation and contextual relevance.

CHAPTER 3: RESEARCH METHODOLOGY

3.1. Research Design

The process of constructing a predictive model for corporate default is a meticulously structured approach that integrates machine learning techniques, financial analysis, and sentiment extraction, ensuring the model's robustness, interpretability, and predictive accuracy. This process is designed to progressively refine and build upon each preceding step, resulting in a comprehensive system that not only forecasts default risk but also provides valuable insights into the factors influencing a company's likelihood of default.

3.1.1. Sentiment Data Processing

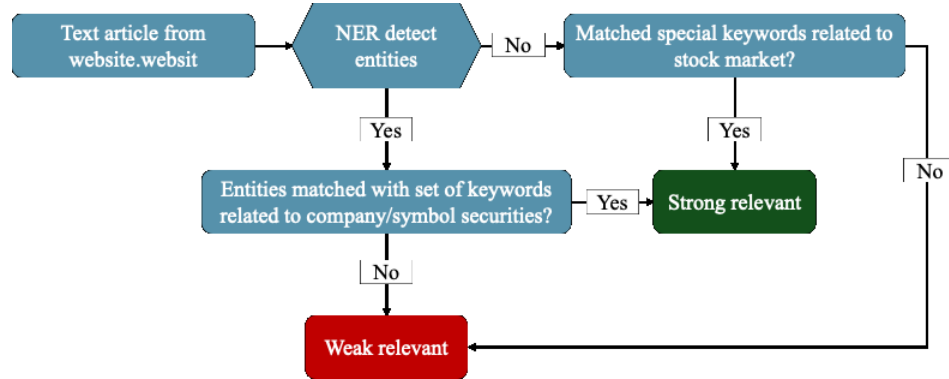
Stock news articles are sourced from reliable financial websites like vnEconomy, Cafef, Vietstock, and VietnamBiz. After data collection, the articles undergo cleaning using the Boilerpipe library (Kohlschütter, 2022), which removes irrelevant content such as advertisements and boilerplate. This ensures that only relevant stock market information is retained for analysis.

Once cleaned, Named Entity Recognition (NER) is used to identify key entities such as company names, stock symbols, and relevant financial terms. Entities such as Hòa Phát (HPG), Vinamilk (VNM), or general terms like "cổ phiếu" (stocks) are identified. These entities help filter articles that are relevant to stock market activities.

(i) Relevance Classification of News Articles

The first step in the process is classifying articles as relevant or irrelevant. Articles that contain recognized stock symbols and keywords related to the stock market are considered relevant and are flagged for further processing. Irrelevant articles, which do not mention stock symbols or financial terms, are immediately discarded as irrelevant.

Figure 3.1: Flowchart for Named Entity Recognition (NER) and Keyword Matching



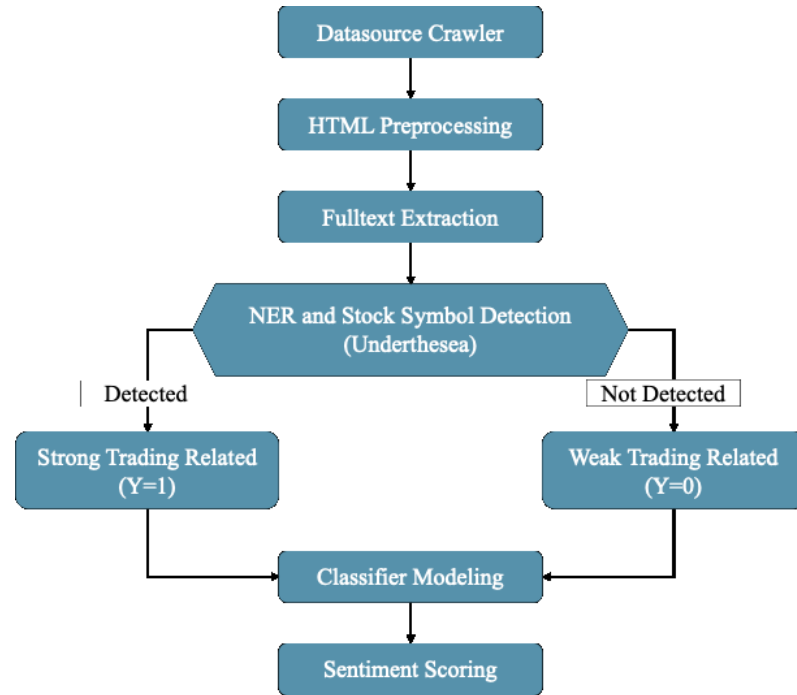
Source: Nhat & Tan (2024)

Once the NER has identified relevant entities and keywords, articles are classified into two categories: strongly relevant (articles that directly pertain to stock market activity) and weakly relevant (articles that mention stocks but do not focus on financial activities). This classification ensures that only the most pertinent articles are analyzed for sentiment.

(ii) Sentiment Scoring with Underthesea

Once the articles are filtered for relevance, they are processed for sentiment analysis. This involves using Underthesea to tokenize the text, assign part-of-speech tags, and detect sentiment-bearing words. For each article, positive terms such as “growth”, “record profits”, and “improvement” are assigned a positive sentiment score of +1. On the other hand, negative terms like “loss”, “decline”, and “setback” are assigned a negative sentiment score of -1. Neutral terms such as “stable” or “no significant change” are classified as neutral sentiment (0).

Additionally, the sentiment analysis process incorporates intensifiers like “very”, “extremely”, and “super” to amplify the sentiment of a phrase, and negation terms like “not” or “never” are used to reverse the sentiment. For example, the phrase “not profitable” will be classified as negative due to the negation.

Figure 3.2: Sentiment Analysis Process with Underthesea

Source: Nhat & Tan (2024)

Once sentiment scores are assigned, the articles are classified into positive, neutral, or negative based on the final score. Articles with scores above a certain threshold are classified as positive (+1), while those below the threshold are classified as negative (-1). Articles with scores close to zero are considered neutral (0).

(iii) Yearly Sentiment Score Calculation

After completing the Named Entity Recognition (NER) and sentiment classification for the stock news articles, the next step involves aggregating the sentiment scores for each stock on an annual basis. This is done by grouping the sentiment scores according to the year and the corresponding stock ticker. Once the articles are grouped, the sentiment scores for each stock in a specific year are collected and averaged to calculate the yearly sentiment score.

This yearly sentiment score is then integrated with the financial report data for the same year. The sentiment score is used to create several key variables, including the Sentiment Score, which represents the average sentiment for the stock

in that year, and the Positive Ratio, Negative Ratio, and Neutral Ratio, which show the proportion of positive, negative, and neutral articles, respectively, among all articles for that stock. In addition, News Volume is calculated as the total number of articles published for the stock in that year. The Sentiment Change reflects the difference in sentiment score compared to the previous year, while News Shock quantifies any significant deviation in sentiment from one year to the next, highlighting major shifts in market perception.

3.1.2. Financial Data Processing

The financial data of companies listed on the HOSE, HNX, and UPCOM exchanges from 2014 to 2024 are sourced from VNStock, a reliable platform providing comprehensive access to financial statements. These financial statements, which include the balance sheet, income statement, and cash flow statement, provide critical insights into the financial health of the companies.

(i) Feature Engineering and Financial Ratio Calculation

In this stage, essential financial ratios are calculated based on the company's financial statements. These ratios serve as the quantitative features used by the model to predict corporate defaults.

The ratios are categorized into several key groups: liquidity ratios, which assess a company's ability to meet short-term obligations (e.g., Current Ratio, Quick Ratio, Working Capital to Total Assets); leverage ratios, which evaluate the company's debt levels (e.g., Debt to Assets, Debt to Equity, Long-Term Debt to Assets); efficiency ratios, which measure how well a company uses its assets to generate revenue (e.g., Receivables Turnover, Inventory Turnover, Asset Turnover); performance ratios, which assess a company's ability to generate profit (e.g., ROA, ROE, Operating Income to Debt); coverage ratios, which measure a company's ability to service its debt (e.g., Interest Coverage); and EBITDA ratios, which provide insights into a company's ability to meet debt obligations from operating profits.

These ratios form the core quantitative features for the model, allowing it to measure the various dimensions of a company's financial health that are predictive of default risk.

(ii) Identifying Signs of Corporate Default

After Feature Engineering and Financial Ratio Calculation, the next step involves identifying signs of corporate default. A company is classified as bankrupt (denoted as 1) if it meets certain criteria. The first step involves assessing the company's operational cash flow (OCF). If a company's OCF is negative, it is flagged as an indicator of financial distress. A further refinement of this metric is made by tracking the company's OCF performance over a rolling three-year period. If a company has experienced negative OCF in at least two of the last three years, it is flagged as a potential risk for default. This method helps to capture ongoing financial struggles that might not be evident from a single year's data but reflect deeper, long-term financial issues.

The second major financial indicator used is the revenue growth, which is measured by the percentage change in revenue from one period to the next. This metric is useful in tracking the overall growth trajectory of a company. A company that consistently shows negative or stagnant revenue growth is considered more vulnerable to default risk. In addition, the study also calculates the compound annual growth rate (CAGR) of revenue over a three-year period, which smooths out short-term fluctuations and provides a more reliable measure of sustained growth or decline. If a company's CAGR is consistently low or negative, it could indicate that the business is struggling to grow at a healthy rate, thus raising the likelihood of financial distress.

Profitability is assessed by analyzing the standard deviation of the company's profit after tax (PAT) over the past three years. High volatility in PAT suggests that a company's financial performance is unstable, which is a crucial indicator of potential default risk. Companies with erratic profit patterns are more likely to face liquidity or solvency issues, which could lead to default. The analysis also includes

a low-risk flag, which combines three other financial ratios: the debt-to-assets ratio, return on assets (ROA), and current ratio. Companies with a debt-to-assets ratio below 50%, an ROA greater than 10%, and a current ratio greater than 1 are categorized as low-risk, which helps identify financially stable businesses less likely to default. Finally, sector-specific default rates are integrated into the model to account for industry-wide economic conditions. These sector default rates provide additional context, reflecting how a company's risk compares to others in the same sector.

(iii) Integration of Financial and Sentiment Data

Once the financial indicators are assessed, the next step is to integrate the sentiment scores calculated earlier for each stock. These sentiment scores, which are derived from relevant stock news articles, are computed by averaging the sentiment scores for all articles related to the stock for a specific year. This average sentiment score is then used to capture the market's overall sentiment towards the company.

To create a complete dataset for the prediction model, the sentiment scores are combined with the financial ratios calculated previously. This comprehensive dataset includes various variables such as the sentiment score, positive ratio, negative ratio, neutral ratio, news volume, sentiment change, and news shock for each stock. By integrating both financial data and sentiment data, the model gains a deeper understanding of the company's financial health and the market's perception, providing a more robust basis for predicting the likelihood of corporate default.

3.1.3. Corporate Default Prediction Pipeline

After integrating financial ratios with sentiment-based variables, the complete dataset is processed through the main prediction pipeline. This pipeline encompasses data preprocessing, model training, and performance validation, forming the core framework for corporate default prediction.

(i) Data Preprocessing

The preprocessing stage ensures that the merged dataset is consistent, reliable, and ready for modeling. Duplicate entries were removed, missing values

were addressed using contextually appropriate imputation methods, and extreme outliers were reviewed to avoid distortions in model learning. This stage ensured that the dataset was both consistent and reliable, forming a solid foundation for subsequent modeling.

The preprocessed dataset was then divided into two parts: a training set, used for model estimation and hyperparameter optimization, and a test set, which was kept strictly separate to allow for an unbiased evaluation of out-of-sample predictive performance.

(ii) Feature Scaling and Transformation

Financial and sentiment features vary considerably in their magnitudes and distributions. To prevent dominant effects from disproportionately large variables, scaling methods were applied, ensuring that all predictors contributed proportionally during training. Additionally, transformations were applied to reduce skewness and stabilize variance in highly non-linear variables. This standardization process made the data more suitable for advanced machine learning algorithms such as gradient boosting.

(iii) Inferring Monotonic Constraints

A distinguishing element of this pipeline is the integration of monotonic constraints into model training. Certain financial ratios follow consistent economic logic in relation to default risk. For example, an increase in leverage ratios such as Debt-to-Assets or Debt-to-Equity typically signals higher risk, while improvements in liquidity and profitability ratios such as Current Ratio or Interest Coverage generally correspond to lower default probabilities.

To empirically verify these expected directions, two complementary approaches were employed:

- SHAP value analysis was used to measure the marginal effect of each feature on predicted default probabilities, comparing bankrupt and non-bankrupt groups.

- Spearman rank correlation was calculated to assess the monotonic association between each feature and default status.

Where SHAP values and correlation signs aligned, a monotonic direction was assigned. In cases of ambiguity, theoretical financial rules were applied as fallback. This hybrid approach ensured that constraints reflected both empirical evidence and domain expertise.

The result was a set of feature-specific monotonic rules as illustrated in Table 3.3, where positive constraints indicate that an increase in the feature corresponds to higher default risk, while negative constraints indicate the opposite relationship.

Table 3.1: Monotonic Constraints and Relationships

Feature	Constraint Direction	Monotonic Relationship
<i>Liquidity Capacity Indicators</i>		
Current Ratio	-1	Increase → ↓ Default Risk
Quick Ratio	-1	Increase → ↓ Default Risk
Working Capital to Total Assets	-1	Increase → ↓ Default Risk
<i>Capital Structure and Leverage Metrics</i>		
Debt to Assets	+1	Increase → ↑ Default Risk
Debt to Equity	+1	Increase → ↑ Default Risk
Equity to Liabilities	-1	Increase → ↓ Default Risk
Long Term Debt to Assets	+1	Increase → ↑ Default Risk
Net Debt to Equity	+1	Increase → ↑ Default Risk
<i>Operational Efficiency Measures</i>		
Receivables Turnover	-1	Increase → ↓ Default Risk
Inventory Turnover	-1	Increase → ↓ Default Risk
Asset Turnover	-1	Increase → ↓ Default Risk
<i>Earnings and Profitability Ratios</i>		

ROA	-1	Increase → ↓ Default Risk
ROE	-1	Increase → ↓ Default Risk
EBIT to Assets	-1	Increase → ↓ Default Risk
Operating Income to Debt	-1	Increase → ↓ Default Risk
Net Profit Margin	-1	Increase → ↓ Default Risk
Gross Margin	-1	Increase → ↓ Default Risk
<i>Coverage Indicators</i>		
Interest Coverage	-1	Increase → ↓ Default Risk
EBITDA to Interest	-1	Increase → ↓ Default Risk
Total Debt to EBITDA	+1	Increase → ↑ Default Risk
<i>Market Sentiment and Information Variables</i>		
Sentiment Score	-1	Increase → ↓ Default Risk
Positive Ratio	-1	Increase → ↓ Default Risk
Negative Ratio	+1	Increase → ↑ Default Risk
Neutral Ratio	0	No Constraint
News Volume	0	No Constraint
Sentiment Change	+1	Increase → ↑ Default Risk
News Shock	+1	Increase → ↑ Default Risk

Source: Statistics from the author

By embedding these constraints directly into gradient boosting models, the pipeline ensures that the learned decision boundaries remain consistent with established financial theory, thereby improving both interpretability and robustness.

(iv) Hyperparameter Optimization with Optuna

Following constraint inference, hyperparameter optimization was conducted using Optuna, a state-of-the-art optimization framework. Specifically, the Tree-structured Parzen Estimator (TPE) algorithm was employed. Unlike conventional grid or random search, TPE constructs a probabilistic model of the objective function, allowing it to prioritize promising regions of the parameter space.

Key parameters optimized included learning rate, maximum tree depth, subsampling ratios, and regularization terms. Optimization was performed independently for three advanced tree-based algorithms including LightGBM, XGBoost, and CatBoost each constrained by the monotonicity rules defined earlier. The optimization objective was to maximize out-of-sample discriminative power, measured by the area under the ROC curve (AUC).

(v) Model Training

After selecting the optimal hyperparameter configurations, models were trained on the full training dataset. The monotonic constraints guided the boosting process, ensuring that the models respected economically consistent relationships while capturing complex non-linearities.

Each algorithm produced a tuned model, yielding a diverse set of predictors that could later be compared for predictive accuracy and stability.

(vi) Model Testing and Validation

The reserved test dataset was employed exclusively for performance evaluation. Predictive accuracy was assessed using metrics such as AUC-ROC, F1-score, precision, recall, and specificity. These measures collectively captured the models' ability to balance false positives and false negatives in default prediction.

Beyond accuracy, rigorous statistical tests were conducted to validate robustness:

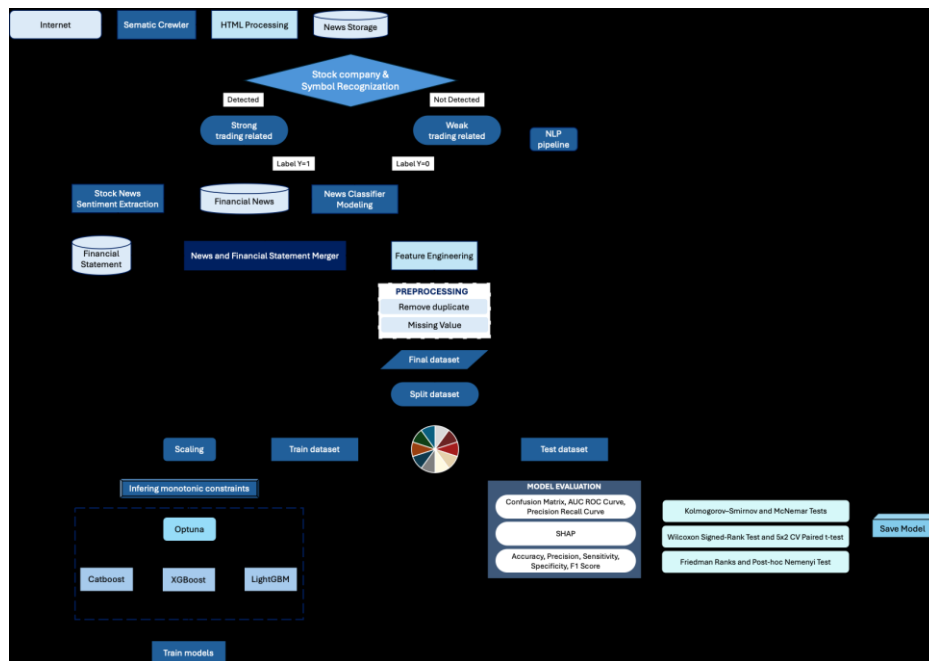
- Kolmogorov–Smirnov and McNemar tests examined distributional differences and classification consistency.
- Wilcoxon Signed-Rank and 5x2 Cross-Validation Paired t-tests evaluated the statistical significance of differences between models.
- Friedman Ranks and post-hoc Nemenyi tests provided a framework for multiple-model comparison.

This layered evaluation strategy ensured that the selected model was not only accurate but also statistically reliable across different validation criteria.

(vii) Model Saving and Deployment

After successful training and evaluation, the final model is saved for future use. The model is serialized using tools like Pickle or Joblib, which enables it to be reloaded and deployed without the need for retraining. Once deployed, the model can be used to make real-time predictions on corporate defaults, and ongoing monitoring is implemented to ensure that the model continues to perform effectively in production.

Figure 3.3: Models Training Process for Predicting Corporate Default



Source: Statistics from the author

3.2. Data collection and processing

3.2.1. Financial Data

This study utilizes financial data derived from the audited financial statements of listed companies in Vietnam between 2014 and 2024 to assess the likelihood of corporate default. The meticulous process of data selection is essential to ensure the quality and reliability of the data, as it directly influences the accuracy and validity of the study's findings. Companies that fail to provide complete financial reports or lack at least two consecutive years of listing on the stock exchange are excluded from the dataset. This exclusion criterion is implemented to enhance the accuracy and feasibility of the analysis, resulting in more reliable

conclusions. The financial accounts used for this study have undergone thorough auditing processes, ensuring the data's accuracy and reliability without significant errors. This research aims to establish a solid foundation for evaluating corporate default risks by strictly adhering to these rigorous data quality standards. Emphasizing certified financial data not only strengthens the credibility of the research but also guarantees that the findings are based on trustworthy and precise financial information.

The dataset analyzed in this study consists of companies from various industries, with 521 companies in the Travel and Leisure sector, 3561 companies in Construction and Materials, 1603 companies in Food and Beverage, 837 companies in Personal & Household Goods, and other sectors including Retail, Real Estate, Industrial Goods & Services, Telecommunications, Cars and Spare Parts, Basic Resources, Medical, Information Technology, Petroleum, Chemicals, Communication, and Electricity, Water & Petroleum Fired. The inclusion of such a wide range of industries is critical to ensure that the study's results are comprehensive and relevant to the diverse economic sectors in Vietnam. Analyzing default rates across these industries allows for a deeper understanding of the specific financial risks each sector faces. Sectors such as Travel and Leisure, Construction and Materials, and Food and Beverage exhibit higher default rates, at 28.6%, 24.07%, and 21.33%, respectively, whereas sectors like Financial Services and Electricity, Water & Gas show significantly lower default rates, at 0% and 4.63%.

Table 3.2: Summary of numbers of enterprises in 10 years– Business sectors

Sector	No Default	Default	Total	Default Rate
Travel and Leisure	372	149	521	28.6%
Construction and Materials	2704	857	3561	24.07%
Food and Beverage	1261	342	1603	21.33%

Personal & Household Goods	686	151	837	18.04%
Retail	259	55	314	17.52%
Real estate	1202	186	1388	13.4%
Industrial Goods & Services	2401	371	2772	13.38%
Telecommunications	79	12	91	13.19%
Cars and spare parts	163	24	187	12.83%
Basic Resources	1045	146	1191	12.26%
Medical	604	70	674	10.39%
Information technology	305	32	337	9.5%
Petroleum	124	12	136	8.2%
Chemicals	717	58	775	7.48%
Communication	370	24	394	6.09%
Electricity, water & petroleum fired	1420	69	1489	4.63%
Financial services	57	0	57	0%
Electricity, water & gas	90	0	90	0%

Source: Statistics from the author

This study places significant emphasis on listed companies, which often encounter distinct challenges compared to larger corporations, underscoring the importance of customized risk assessment methods that address the particular vulnerabilities of these businesses. Furthermore, the analysis of default rates over the period from 2014 to 2024 reveals a slight decrease in default rates, from 16.63% in 2014 to 14.36% in 2024, indicating an improvement in the ability of listed companies to manage and service their debts over this period.

Table 3.3: Separate default and non-default enterprises

Year	No Default	Default	Total	Default Rate (%)
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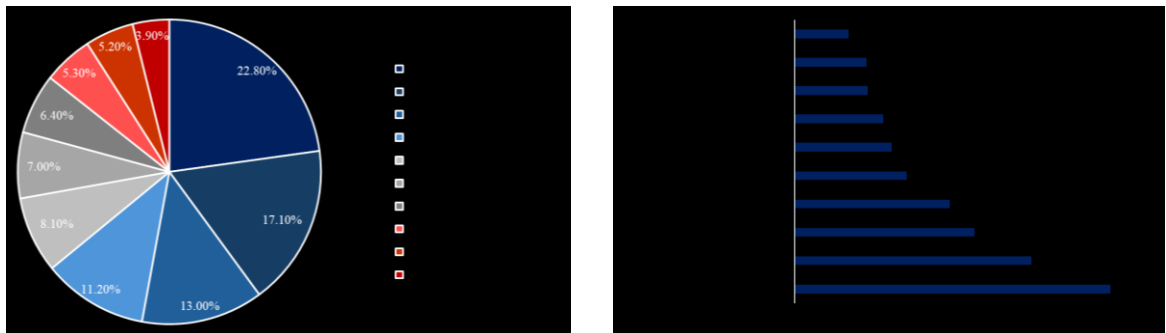
2014	1043	208	1251	16.63%
2015	1093	220	1313	16.76%
2016	1148	225	1373	16.39%
2017	1160	223	1383	16.12%
2018	1189	221	1410	15.67%
2019	1188	219	1407	15.57%
2020	1154	200	1354	14.77%
2021	1224	213	1437	14.82%
2022	1236	212	1448	14.64%
2023	1232	207	1439	14.38%
2024	1217	204	1421	14.36%

Source: Statistics from the author

3.2.2. Non-financial Data

The non-financial layer is a longitudinal Vietnamese news corpus crawled from 2014 through 2024, targeting finance and stock-market sections only. Over this full horizon the crawler collected more than 500,000 articles, which corresponds to roughly 45,000 items per year on average. Sources span both national business portals and specialist securities publishers and are summarized in Figure NF1; we do not repeat outlet shares in the text to avoid redundancy.

Figure 3.4: Data source description for the news classification model



Source: Statistics from the author

Collection relies on a semantic crawler that respects robots rules and paginated archives, storing URL, outlet, timestamp, headline, and full body. HTML

is cleaned with a boilerplate-removal routine that preserves numerals and punctuation carrying sentiment cues, while canonicalization and near-duplicate hashing suppress syndicated duplicates. Timestamps are normalized to Vietnam local time and later aligned at the year level for integration with financial statements. To link articles with listed firms, a master dictionary of company names, aliases, and tickers is maintained for HOSE, HNX, and UPCoM. For illustration, the dataset includes companies such as FPT Corporation (FPT) and Mobile World Investment Corporation (MWG) from HOSE, Saigon–Hanoi Commercial Joint Stock Bank (SHB) from HNX, and Airports Corporation of Vietnam (ACV) from UPCoM. These examples highlight the industry diversity in the dataset and underscore the importance of precise entity recognition. Vietnamese-specific preprocessing is performed with Underthesea for tokenization and, when needed, POS tagging, while a named-entity layer combines dictionary lookup, fuzzy matching, and market keywords to detect issuer mentions and trading context. Articles without mapped entities or stock-domain cues are filtered out, leaving only the market-related subset.

For this subset, a rule-based sentiment engine enhanced by Underthesea assigns labels of negative (-1), neutral (0), or positive (1). The engine captures clause-level negation, intensity modifiers, uncertainty cues, contrastive connectors, and numeric expressions such as percentage changes, index-point moves, limit-up/limit-down events, and KPI growth statements. Debug traces of matched phrases and numbers are preserved to ensure auditability. Finally, sentiment scores are aggregated at the ticker–year level, producing a non-financial data panel that is subsequently merged with the annual financial panel. This merged dataset forms the foundation for feature construction and corporate default prediction modeling.

3.3. Selection of input variables in the default prediction model

A principled selection of input variables is essential for credible default prediction, especially when the final model enforces monotonic constraints on economically ordered signals. According to Zhao et al. (2024), contemporary

surveys of bankruptcy and financial distress research consistently show that accounting ratios remain the backbone of predictive performance while complementary information from text and other alternative data raises early-warning power and stability out of sample. Alvi et al (2024) argues that models built on grouped ratio families, augmented with modern machine learning, constitute the dominant and most reproducible design across jurisdictions and firm sizes.

Typically, default prediction models categorise independent variables into four basic groups of financial ratios.

(i) Liquidity Capacity Indicators

Liquidity indicators are fundamental to assessing the short-term solvency of listed firms. The Current Ratio, Quick Ratio, and Working Capital to Total Assets capture the ability of corporations to meet imminent liabilities with liquid resources. According to Abdelkader and Wahba (2024), working capital scaled by assets is one of the most consistent early predictors of financial distress, highlighting its role in identifying weakening solvency structures in listed entities. Charalambakis and Garrett (2019) argue that liquidity indicators remain indispensable across credit risk models because declining liquidity often precedes covenant breaches and eventual default. Hernandez Tinoco and Wilson (2013) emphasize that liquidity ratios, when combined with profitability and leverage, significantly improve bankruptcy prediction by capturing the firm's resilience against cash flow shortfalls.

(ii) Capital Structure and Leverage Metrics

The configuration of debt and equity financing critically shapes the probability of default. Ratios such as Debt to Assets, Debt to Equity, Equity to Liabilities, and Long-Term Debt to Assets measure the structural vulnerability of corporations. Gallucci et al. (2023) argue that leverage ratios exert a strong influence on default prediction models, with higher indebtedness translating directly into higher insolvency risk. According to Zhao et al. (2024), leverage variables consistently appear among the most selected predictors in modern machine learning

approaches, demonstrating their universal significance across industries. Charalambakis and Garrett (2019) further assert that leverage indicators intensify the impact of liquidity shortages, creating a compounding effect that accelerates financial distress.

(iii) Operational Efficiency Measures

Efficiency ratios provide insight into the capacity of firms to manage assets and operations effectively. Indicators such as Receivables Turnover, Inventory Turnover, and Asset Turnover are closely linked to cash cycle management and resource allocation. Abdelkader and Wahba (2024) argue that efficiency, particularly asset turnover, is a statistically significant negative predictor of corporate distress, reflecting that more effective resource utilization reduces failure risk. W. Wang (2024) shows that efficiency measures complement liquidity and profitability in predicting default by identifying weaknesses in operations before they manifest in earnings decline. Halteh (2025) emphasizes that asset turnover provides distinct information about managerial quality and operational discipline, which are critical for sustaining long-term solvency.

(iv) Earnings and Profitability Ratios

Profitability is among the most powerful determinants of corporate default. Ratios including Return on Assets (ROA), Return on Equity (ROE), EBIT to Assets, Operating Income to Debt, Net Profit Margin, and Gross Margin describe the ability of firms to generate sustainable earnings relative to assets, equity, and obligations. Abdelkader and Wahba (2024) argue that EBIT to assets is a leading negative predictor of default, capturing the firm's fundamental earning capacity. According to Charalambakis and Garrett (2019), profitability ratios improve classification of distressed firms, reflecting their central role in determining whether companies can absorb external shocks. Agrawal and Maheshwari (2019) highlight that profitability interacts with liquidity and leverage, as declining margins simultaneously weaken internal funding and exacerbate debt servicing burdens.

(v) Coverage Indicators

Coverage ratios translate operating earnings into capacity for meeting fixed charges. The Interest Coverage Ratio is a widely recognized predictor of default, as it indicates how comfortably firms can service interest expenses. Aoki and Hosonuma (2004) argues that interest coverage is the most important variable for bankruptcy prediction in decision-tree models, highlighting its dominant role relative to other ratios. Charalambakis and Garrett (2019) support this view, noting that interest coverage adds incremental explanatory power beyond profitability and leverage, since creditors explicitly monitor such metrics to determine early intervention. Ma et al. (2025) document that lenders rely on volatility in coverage measures as a proxy for default likelihood, which aligns with the contractual use of minimum coverage covenants. EBITDA-related measures focus on earnings capacity before financing and taxation, reflecting sustainable cash flows available for debt servicing. Indicators such as EBITDA to Interest, Total Debt to EBITDA, and Net Debt to Equity are widely used in lending agreements. Hollander and Verriest (2016) argue that debt-to-EBITDA covenants are among the most prevalent and homogeneous financial tests in loan contracts, underscoring their predictive reliability. Demerjian and Owens (2016) show that EBITDA-based metrics are highly informative in forecasting covenant violations, which often precede technical default. According to Roberts (2015), EBITDA-related ratios act as dynamic control rights, allowing lenders to renegotiate or impose restrictions as borrower health deteriorates.

(vi) Market Sentiment and Information Variables

Beyond accounting data, sentiment extracted from unstructured sources provides forward-looking signals of financial distress. Measures such as Sentiment Score, Positive, Negative, and Neutral Ratios, News Volume, Sentiment Change, and News Shock enrich the predictive space by capturing market perception and narrative shifts. D. N. Nguyen et al. (2022a) argues that integrating sentiment with financial ratios significantly improves bankruptcy prediction by reflecting investor expectations ahead of accounting disclosures. According to Audrino et al. (2020),

news sentiment shapes short-term market volatility, serving as an early proxy for deteriorating conditions. Roeder et al. (2020) show that topic-level news indicators improve monitoring of credit risk by banks, while Silver (2024) argue that sentiment amplifies market dynamics and default risk through expectation channels.

Table 3.4: Summary of variables

Variable	Group of Financial Indicator	Expected Effect	Source
<i>Dependent variable</i>			
DEFAULT			(Agrawal & Maheshwari, 2019; Andresson & Lukason, 2024; Avgeri & Psillaki, 2024; Bonacorsi et al., 2024; Gallucci et al., 2023; Mehmood & De Luca, 2023; Yin et al., 2024)
<i>Independent variable</i>			
Current Ratio	Liquidity	-	(Gallucci et al., 2023; López-Sánchez et al., 2022; Mehmood & De Luca, 2023; W. Wang & Guedes, 2025)
Quick Ratio	Liquidity	-	(Avgeri & Psillaki, 2024; Gallucci et al., 2023; Mehmood & De Luca, 2023a; Halteh, 2025; Peliu, 2024)
Working Capital	Liquidity	-	(Agrawal & Maheshwari,

to Total Assets			2019; Desai & Mehta, 2024; Gallucci et al., 2023; Lyngstadaas, 2020)
Debt to Assets	Leverage	+	(Coimbra, 2020; Gallucci et al., 2023; Zeitun & Refai, 2017; Alvi et al., 2024)
Debt to Equity	Leverage	+	(Gallucci et al., 2023)
Equity to Liabilities	Leverage	-	(Abdelkader & Wahba, 2024)
Long Term Debt to Assets	Leverage	+	(Duvall & Rathinasamy, 1993; Gallucci et al., 2023; Jiang et al., 2025; Nakagawa & Yamanaka, 2025)
Net Debt to Equity	Leverage	+	(Gallucci et al., 2023; Yin et al., 2024)
Receivables Turnover	Efficiency	+	(X. Yang, 2023)
Inventory Turnover	Efficiency	-	(Xie & Xie, 2025)
Asset Turnover	Efficiency	-	(Ashik-Uz-Zaman et al., 2025; Gabrielli & Greco, 2023; Gallucci et al., 2023; Peliu, 2024)
ROA	Profitability	-	(Gallucci et al., 2023; Peliu, 2024; Jiang et al., 2025; Wang & Guedes,

			2025)
ROE	Profitability	-	(Gallucci et al., 2023; Jiang et al., 2025; Peliu, 2024; W. Wang & Guedes, 2025)
EBIT to Assets	Profitability	-	(Abdelkader & Wahba, 2024)
Operating Income to Debt	Profitability	-	(Nakagawa & Yamanaka, 2025)
Net Profit Margin	Profitability	-	(Luu Thu, 2023)
Gross Margin	Profitability	-	(Lyócsa et al., 2022)
Interest Coverage	Coverage	-	(Arhinful & Radmehr, 2023; Giráldez-Puig & Samaniego-Medina, 2025)
EBITDA to Interest	Coverage	-	(Wang et al., 2024)
Total Debt to EBITDA	Coverage	+	(Gallucci et al., 2023; Wang et al., 2024)
Sentiment Score	Sentiment	-	(Huang et al., 2023; Isakin & Pu, 2023; Liebmann et al., 2016; Smales, 2016)
Positive Ratio	Sentiment	-	(S. Li et al., 2022)
Negative Ratio	Sentiment	+	(Ahmad et al., 2016; C. Y. H. Chen et al., 2022; Divernois & Filipović, 2024; Du et al., 2022;

			Gao et al., 2022; Gu et al., 2022; Jeon et al., 2022; S. Liu & Han, 2020; Shen et al., 2022; W. Wang, 2024)
Neutral Ratio	Sentiment		(Audrino et al., 2020; Colasanto et al., 2022; Divernois & Filipović, 2024; Du et al., 2022; Varghese & Mohan, 2023)
News Volume	Sentiment		(Ashwin, 2024; Dang et al., 2019; Jeon et al., 2022; M. Li et al., 2024; Mizuno et al., 2017; Tsai et al., 2016)
Sentiment Change	Sentiment	+	(Dergiades, 2012; Kim & Ryu, 2022; Smales, 2016; W. Wang, 2024; X. Yang, 2023)
News Shock	Sentiment	+	(Benhima & Cordonier, 2022; Dvorkin et al., 2020; Fong, 2021; Metiu & Prieto, 2023; Sayeed & Islam, 2025)

Source: Author's own synthesis

3.4. Models for predict the probability of default

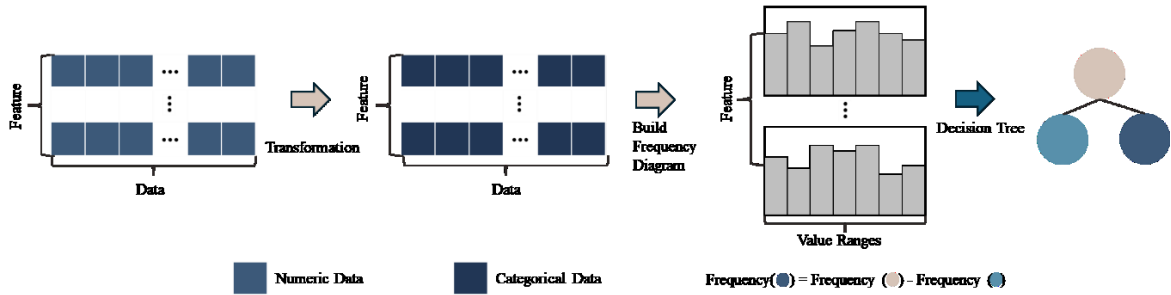
3.4.1. *LightGBM*

Light Gradient Boosting Machine (LightGBM) represents an advanced implementation of gradient boosting that leverages decision tree learning. It is designed to provide exceptional computational efficiency and predictive accuracy,

making it particularly well-suited for large-scale and complex datasets. Ke et al. (2017) introduced two key innovations in LightGBM: Gradient-based One Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). These innovations significantly improve training speed and scalability while preserving performance that is virtually indistinguishable from traditional GBDT implementations.

A major advancement of the algorithm lies in its data sampling strategy. The GOSS method retains observations with the largest absolute gradients while randomly sampling a subset from those with smaller gradients. By prioritizing the most informative instances, GOSS reduces the number of observations processed at each iteration. This sampling approach includes a correction factor to adjust for the under-sampling of small-gradient instances in the computation of information gain, ensuring that the estimates remain aligned with the original data distribution. Figure 3.5 illustrates this process, showing how the data is transformed and classified into distinct intervals based on frequency, ultimately aiding the construction of the decision tree.

Figure 3.5: Decision Tree Algorithm Based on Frequency Histogram

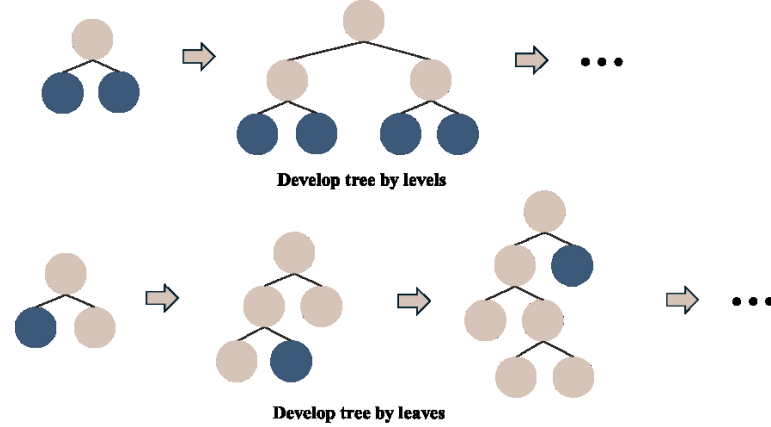


Source: (Liang et al., 2020)

Another significant contribution of LightGBM is Exclusive Feature Bundling (EFB), which reduces the dimensionality of the feature space in a near-lossless manner. Mutually exclusive features—those that rarely take nonzero values simultaneously—are grouped into composite features. Although determining the optimal bundling arrangement is NP-hard, a greedy approximation method achieves highly effective reduction. This bundling not only improves memory efficiency but also accelerates computation, particularly when building histograms during tree

construction, without compromising predictive accuracy. Figure 3.6 demonstrates two distinct tree construction methods: level-wise and leaf-wise. The algorithm splits the leaf offering the maximal reduction in loss, which provides a clear view of how LightGBM develops trees in a leaf-wise manner.

Figure 3.6: Level-wise and Leaf-wise Tree Construction Methods



Source: (Liang et al., 2020)

From a mathematical perspective, LightGBM optimizes the following regularized objective at each boosting iteration t :

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t)$$

where $\ell(\cdot)$ is a differentiable loss and $\Omega(f_t)$ denotes the regularization applied to the tree structure. Leaf growth proceeds in a leaf-wise manner: at each step the algorithm chooses the leaf that offers the maximal reduction in loss to split, subject to depth constraints that guard against overfitting. Gradient descent, augmented by GOSS and EFB, guides this process efficiently, integrating data reduction and feature bundling into a coherent optimization framework.

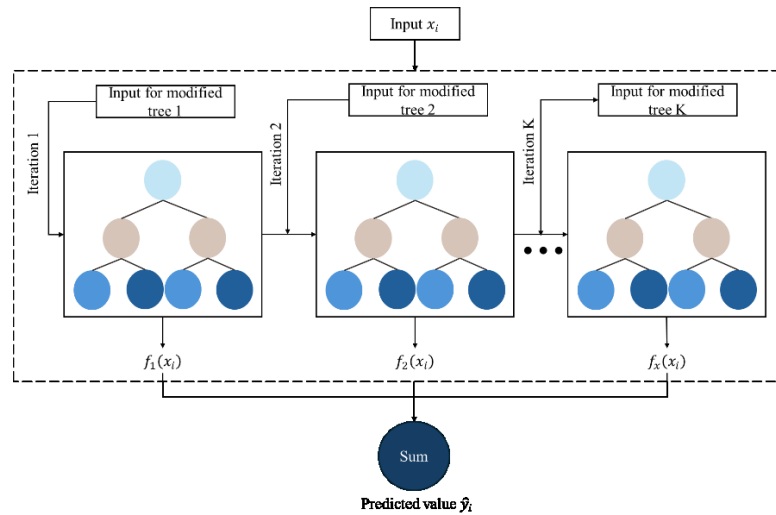
3.4.2. XGBoost

Extreme Gradient Boosting, commonly known as XGBoost, is a scalable and accurate implementation of gradient boosting that integrates explicit regularization to control model complexity and reduce overfitting. The algorithm begins with an initial prediction for each observation, often the average of the target variable, and

then improves this prediction in a stage-wise manner by adding trees that correct the remaining errors. At each boosting round, the algorithm computes the first-order gradient and the second-order Hessian of the chosen loss with respect to the current prediction. These derivatives provide informative directions and curvatures that guide both split finding and weight updates, which is central to the efficiency and stability of the method.

Tree construction uses the calculated gradients and Hessians to grow a new regression tree that best reduces the objective. Candidate splits are evaluated by their expected loss reduction, and the tree expands along those splits that produce the largest gains. After the tree is grown, XGBoost applies pruning through a complexity threshold that removes branches whose contribution to loss reduction is insufficient. This combination of informed growth and post-growth pruning curbs unnecessary depth and helps the model generalize beyond the training data. Figure 3.7 illustrates the iterative process of adding trees and correcting errors, showing how each iteration refines the prediction.

Figure 3.7: Schematic diagram of the XGBoost regression tree model



Source: (Zou et al., 2022)

Regularization is embedded directly in the learning framework. XGBoost penalizes the number of leaves and the magnitude of leaf weights by combining L1 and L2 penalties, which shrinks parameters toward zero and discourages overly

complex trees. The overall objective couples data fit and model parsimony. A common formulation at boosting round t is

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \sum_{k=1}^t \Omega(f_k)$$

where $\ell(\cdot)$ is the loss function and $\Omega(f)$ is a regularization term that typically takes the form

$$\Omega(f) = \gamma T + \frac{\lambda}{2} \sum_{j=1}^T w_j^2 + \alpha \sum_{j=1}^T |w_j|$$

With T denoting the number of leaves, w_j the leaf weights, and γ, λ, α nonnegative regularization parameters. Optimization proceeds by minimizing $\mathcal{L}^{(t)}$ at each stage, yielding a sequence of trees whose additive predictions converge toward the risk minimizer under the specified loss. Through this design, XGBoost unites second order optimization, informed pruning, and structured regularization to deliver high accuracy with strong control of overfitting while preserving computational efficiency for large and complex datasets (Chen and Guestrin, 2016).

3.4.3. *CatBoost*

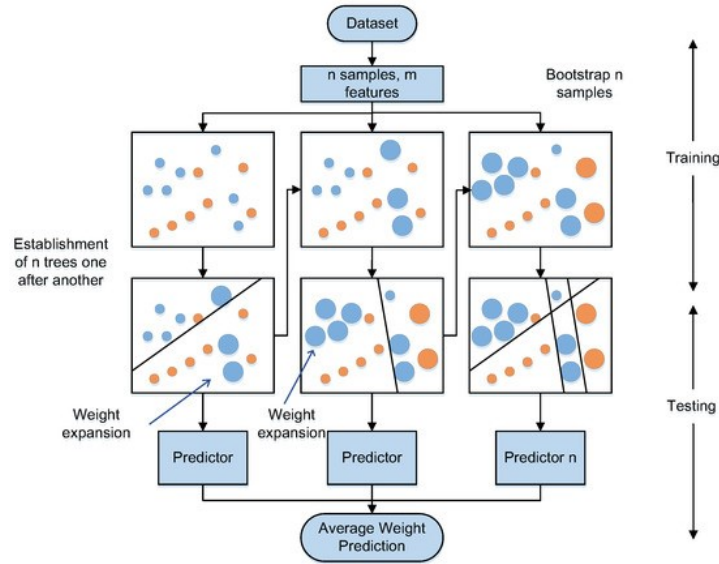
CatBoost stands out as a gradient boosting framework that is specifically tailored to handle categorical predictors in a principled and robust way. According to Prokhorenkova et al. (2018), traditional gradient boosting models are susceptible to a special kind of target leakage whereby a model's predictions for training examples differ in distribution from unseen test examples. To address this, the authors introduce the method of ordered boosting, which employs random permutations of the training data and restricts each update to information derived only from earlier observations in that permutation. By applying this ordering principle, CatBoost effectively eliminates the prediction shift caused by leakage (Prokhorenkova et al., 2018).

The categorical feature encoding in CatBoost also benefits from the ordered framework. According to Hancock (2020), CatBoost employs ordered target

statistics derived from permutations to compute target-based encodings without introducing leakage. For each categorical value, the algorithm replaces it with the average of prior target values observed in the current permutation, smoothed toward an overall prior. This allows encoded categories to evolve as more data accumulates, thereby preventing overfitting that arises from naive target encoding methods.

Figure 3.8 illustrates the process of how CatBoost sequentially builds trees based on randomized data inputs, ensuring that each new tree corrects errors from prior iterations. This process allows the model to adapt and minimize bias through consistent updates while maintaining its predictive integrity.

Figure 3.8: The structure of the CatBoost algorithm



Source: (H. Yang et al., 2023)

CatBoost optimizes a loss comprised of the prediction error and an implicit regularization term. Geeitha et al. (2024) provide formulas illustrating this objective. The gradient with respect to a categorical encoding weight $U_{i,j}$ is expressed as

$$\frac{\partial L}{\partial F(y_i)} \cdot \frac{\partial F(y_i)}{\partial U_{i,j}}$$

and the Hessian accounts for second-order effects as

$$\frac{\partial^2 L}{\partial F(y_i)^2} \cdot \left(\frac{\partial F(y_i)}{\partial U_{i,j}} \right)^2 + \frac{\partial F(y_i)}{\partial U_{i,j}} \cdot \frac{\partial^2 F(y_i)}{\partial U_{i,j}^2}$$

This allows splitting decisions that minimize an objective of the form

$$obj = \frac{(\sum_{j \in leaf(u)} \partial L(x, F(y_i)) / \partial F(x_i))^2}{\sum_{j \in leaf(u)} \partial^2 L(x, F(y_i)) / \partial F(x_i)^2 + \alpha}$$

where α is a regularization constant. This formula elegantly balances information about gradients and Hessians to select splits that most effectively reduce loss while controlling complexity.

3.5. Methods for selecting a model to predict default probability

3.5.1. Testing and evaluating the model on out-of-sample data

Out-of-sample testing plays a crucial role in validating predictive models by evaluating their performance on unseen data. This approach involves using a separate dataset, not included during the training phase, to assess how well the model generalizes to novel, real-world situations. It is vital for detecting issues like overfitting or underfitting, helping optimize the model before deployment.

Firstly, out-of-sample testing helps evaluate a model's ability to generalize beyond the training data. While training a model on a specific dataset may lead to high performance on that data, out-of-sample testing ensures the model applies its knowledge to new data, reflecting its ability to recognize fundamental patterns instead of memorizing the training set.

Secondly, out-of-sample testing is key in detecting overfitting, which occurs when a model is too tailored to the training data and performs poorly on unseen data. By comparing the model's performance on both the training and test datasets, researchers can identify overfitting and adjust the model to improve generalization.

Thirdly, the insights gained from out-of-sample testing guide model refinement. Adjustments such as hyperparameter tuning, feature engineering, and algorithm selection help enhance the model's ability to handle diverse, real-world conditions, improving its overall performance.

Lastly, out-of-sample testing increases the reliability of the model's predictions by validating its performance in environments that simulate real-world scenarios. This process reduces the risk of deploying a model that only performs well under controlled conditions, ensuring it is a trustworthy tool for decision-making.

3.5.2. Criteria to measure the model's ability to predict default risk

To evaluate the effectiveness of predictive models for calculating corporate default probability, various metrics such as the Confusion Matrix, Accuracy ratio, Sensitivity ratio, Specificity ratio, AUC, and F1-score are used. These metrics provide a comprehensive assessment of the model's performance, highlighting its strengths and weaknesses in terms of prediction accuracy.

(i) The confusion matrix

The Confusion Matrix plays a central role in evaluating classification models for corporate default prediction by illustrating how well a model distinguishes between default and non-default cases. Through the distribution of True Positives, True Negatives, False Positives, and False Negatives, it reveals the trade-offs in prediction accuracy. A high number of False Negatives signals that the model fails to capture the complexities of financial risk, while many False Positives suggest excessive sensitivity to certain factors. Interpreting these outcomes supports adjustments to thresholds or features, making the Confusion Matrix a vital tool for refining models in both research and practical financial applications.

Table 3.5: Confusion Matrix

	Predicted class		
Actual class	<i>Classes</i>	<i>Non-default</i>	<i>Default</i>
	<i>Non-default</i>	True Negative (TN)	False Positive (FP)
	<i>Default</i>	False Negative (FN)	True Positive (TP)

Source: Compiled from the author

The Confusion Matrix is a square matrix that lays out the classification outcomes in a structured manner, capturing both the successes and failures of a

model's predictions. It is divided into four quadrants, each representing a different outcome of the model's classification process.

Table 3.6: Metrics for each class in the Confusion Matrix

True Positive (TP)	The model accurately predicts a default. This is critical in credit risk management, allowing financial organizations to proactively mitigate potential losses.
True Negative (TN)	The model correctly predicts non-default. While less emphasized, TN is important to avoid overestimating risk, preventing unnecessary restrictions on financing and growth.
False Positive (FP)	The model incorrectly predicts a default for a stable firm. These errors can lead to unjust consequences, such as higher borrowing costs or credit denial for healthy firms.
False Negative (FN)	The model fails to predict a default, misclassifying a high-risk firm as low-risk. FN errors are the most dangerous, leaving financial institutions exposed to unforeseen losses.

Source: Compiled from the author

(iii) Other performance metrics: Accuracy, Precision, Sensitivity, Specificity, F1 score, AUC

Based on the four fundamental indices previously discussed, researchers have developed a comprehensive set of metrics to evaluate the reliability and effectiveness of corporate default prediction models. These measures not only assess the precision of the model's forecasts but also offer a comprehensive perspective on any deficiencies in the categorisation procedure.

Table 3.7: Groups of indicators to evaluate the reliability of a model

Indicators	Definition	Formula
Accuracy	Denotes the ratio of accurately	Accuracy =

	predicted outcomes (including both true positives and true negatives) to the total number of instances.	$\frac{(TN + TP)}{(TN + FN + TP + FP)}$
Sensitivity	Evaluates the model's ability to precisely identify real instances of corporate default.	$\text{Sensitivity} = TP/(TP+FN)$
Precision	Calculates the accuracy of default forecasts by determining the ratio of correctly identified bankrupt firms to the total number of corporations anticipated to go bankrupt.	$\text{Precision} = \frac{TP}{(TP + FP)}$
F1 score	Calculates a unified metric of Precision and Sensitivity that considers both false positives and false negatives.	$F1 \text{ score} = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}$

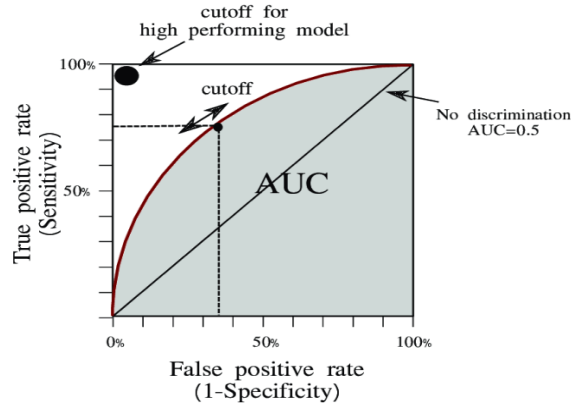
Source: Nguyen, M. N., & Ngo, H. K. D. (2024)

In binary classification tasks such as credit risk assessment and medical diagnosis, it is essential to use diverse evaluation metrics to obtain a comprehensive view of model performance. The Receiver Operating Characteristic (ROC) curve illustrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) across thresholds, while the Area Under the Curve (AUC) quantifies the model's overall discriminative ability. As a foundational metric, AUC is particularly useful in settings with class imbalance, yet its interpretation should be complemented with measures such as Precision, Recall, and the F1-score to evaluate prediction accuracy for minority classes.

Beyond its role as a performance indicator, AUC connects to broader statistical measures such as the Gini coefficient and the Mann–Whitney U test, underscoring its theoretical significance. To ensure generalizability and robustness, AUC should be assessed in combination with complementary metrics and cross-

validation procedures, followed by continuous monitoring as models are deployed. This integrated evaluation framework allows researchers and practitioners to maintain reliable and accurate predictions when applying models to evolving real-world datasets.

Figure 3.9: Simulation of AUC



Source: Compiled from the author

3.6. Nonparametric and Paired Statistical Tests

3.6.1. Kolmogorov–Smirnov (two-sample, distributional separation)

According to Bradley (2013), KS offers a nonparametric equivalence check between predictive-score distributions that maps coherently to ROC characteristics, aiding decision-threshold design in credit risk evaluation.

Formula:

$$D_{n,m} = \sup_x |F_n(x) - G_n(x)|$$

Reject for large $D_{n,m}$ using the exact or asymptotic reference distribution.

3.6.2. McNemar Test (paired error proportions on the same cases)

According to McNemar (1947), equality of error rates for two classifiers evaluated on identical observations should be tested with discordant counts only, avoiding dependence bias from shared cases.

Formula:

$$\chi^2 = \frac{(b-c)^2}{b+c} \text{ (no correction)}$$

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c} \text{ (continuity-corrected)}$$

3.6.3. Wilcoxon Signed-Rank (paired metric differences)

García et al. (2010) argue that Wilcoxon is appropriate for fold-level paired differences that violate normality, offering power and robustness without parametric assumptions. Derrac et al. (2011) document its routine use in ML comparisons over repeated runs.

Formula:

Let d_i be non-zero paired differences; rank d_i to R_i .

$$W^+ = \sum_{d_i > 0} R_i$$

$$T = \sum_i \text{sgn}(d_i) R_i$$

Use the exact distribution (small n) or normal approximation.

3.6.4. 5×2 CV Paired t-Test (two algorithms, variance-aware)

According to Dietterich (1998), five repetitions of two-fold CV reduce overlap variance and yield a stable paired comparison for exactly two algorithms; the test statistic aggregates split-wise differences with an appropriate variance estimator. Subsequent work reinforces the design and proposes related F-variants.

Let $d_{i,j}$ be the performance difference for repetition $i \in \{1, \dots, 5\}$ and half $j \in \{1, 2\}$.

$$\bar{d}_i = \frac{d_{i,1} + d_{i,2}}{2}$$

$$s_i^2 = (d_{i,1} - \bar{d}_i)^2 + (d_{i,2} - \bar{d}_i)^2$$

$$t = \frac{\sum_{i=1}^5 d_{i,1}}{\sqrt{\frac{1}{5} \sum_{i=1}^5 s_i^2}}$$

with reference to the t distribution as specified in Dietterich (1998).

3.6.5. Friedman Ranks (omnibus, kkk models over NNN matched datasets)

According to García et al. (2010), Friedman supplies a distribution-free omnibus test for multiple-algorithm comparisons, with power improving as N

increases. Derrac et al. (2011) provide a practical tutorial for experimental design and reporting.

Formula:

$$\text{Average rank for model } j: R_j = \frac{1}{N} \sum_{i=1}^N r_{ij}$$

$$\chi_F^2 = \frac{12N}{k(k+1)} \left(\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right), F_F = \frac{(N-1)\chi_F^2}{N(k-1)-\chi_F^2}$$

3.6.6. *Post-hoc Nemenyi (all-pairs after Friedman)*

According to García et al. (2010), when no single control model is designated, the Nemenyi procedure compares all pairs of average ranks under familywise control; Derrac et al. (2011) recommend reporting critical differences alongside rank plots to localize effects transparently.

Formula:

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \text{ declare } |R_j - R_l| > CD \text{ as significant.}$$

CONCLUSION OF CHAPTER 3

This chapter presented the research methodology for developing a predictive model for corporate default, combining financial analysis, sentiment extraction, and machine learning techniques. Data was collected from financial statements, news articles, and market data, with sentiment analysis enriching the dataset. Key financial ratios, including the Current Ratio, Debt to Assets, and ROA, were used in feature engineering to assess default risk.

The model identified companies at risk of default using criteria like negative net income and consecutive losses. Data preprocessing ensured clean, consistent data, while monotonic constraints based on SHAP values and Spearman correlation aligned predictions with financial principles.

Advanced tree-based models, such as XGBoost, LightGBM, and CatBoost, were employed, with hyperparameter tuning using Optuna. Model performance was evaluated through metrics like the Confusion Matrix, AUC-ROC, Precision, Recall, and F1-Score, with statistical tests confirming robustness.

The methodology integrates machine learning and financial analysis to ensure a robust and interpretable model for corporate default prediction. Chapter 4 will present the empirical results and further validation of the model's effectiveness.

CHAPTER 4: EMPIRICAL RESULTS AND DISCUSSION

4.1. Descriptive statistics of variables

This section presents descriptive statistics of the variables used in the study, including the number of observations, mean, standard deviation, minimum, and maximum values. These indicators provide an overall understanding of the dataset. Table 4.1 The descriptive statistics are based on 16,417 observations collected from listed firms on the Vietnamese stock exchanges (HOSE, HNX, and UPCoM) over the period 2014–2024.

Table 4.1: Descriptive statistics of variables

Variable	Number of observations	Mean value	Standard deviation	Minimum value	Maximum value
Current Ratio	16.417	1,019157	0,038528	0,75098	1,536734
Quick Ratio	16.417	1,89623	0,096333	0,677571	2,909117
Working Capital to Total Assets	16.417	0,017324	0,035253	-0,29564	0,349269
Debt to Assets	16.417	2,380184	0,620828	0,60511	3,318568
Debt to Equity	16.417	2,358761	0,799255	0	3,893338
Equity to Liabilities	16.417	0,446402	0,231233	0	1,652589
Long Term Debt to Assets	16.417	0,700946	0,360632	0	1,097364
Receivables Turnover	16.417	1,046814	0,125481	0	1,60241
Inventory Turnover	16.417	0,000824	0,028376	0	1,243807
Asset Turnover	16.417	0,975276	0,109984	0	1,31243

ROA	16.417	0,752269	0,305895	0	1,056302
ROE	16.417	0,768854	0,320221	0	1,244979
EBIT to Assets	16.417	0,729973	0,336378	0	1,060515
Operating Income to Debt	16.417	0,348991	0,240406	0	1,281536
Net Profit Margin	16.417	0,758071	0,312436	0	1,570031
Gross Margin	16.417	0,879301	0,211852	0	1,022798
Interest Coverage	16.417	0,00209	0,048363	0	1,532497
EBITDA to Interest	16.417	0,00367	0,084473	0	2,920886
Total Debt to EBITDA	16.417	1,672786	0,827123	0	5,066852
Net Debt to Equity	16.417	1,481815	0,71398	-0,32302	2,768797
Sentiment Score	16.417	0,139494	0,178673	0	0,693147
Positive Ratio	16.417	0,249504	0,109416	0,14842	0,578858
Negative Ratio	16.417	0,235724	0,114451	0,144966	0,576613
Neutral Ratio	16.417	0,353235	0,122044	0,058269	0,494696
News Volume	16.417	6,333866			
Sentiment Change	16.417	0,076441	0,147492	0	1,095273
News Shock	16.417	0,036335	0,089405	0	0,904623

Default	16.417	0,155814	0,36269	0	1
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Source: Author's calculation

The Current Ratio exhibits a mean of 1.0192 with a standard deviation of 0.0385. This indicates relatively low variation across listed firms, with the lowest observation recorded at 0.7509 and the maximum at 1.5367. The Quick Ratio presents a higher mean of 1.8962 and a standard deviation of 0.0963, reflecting greater dispersion in firms' liquid asset capacity. The values range from 0.6776 to 2.9091, showing significant differences in the ability to cover short-term liabilities without relying on inventory. Similarly, Working Capital to Total Assets averages 0.0173, with considerable variation (standard deviation of 0.0353) and extremes ranging from -0.2956 to 0.3493, highlighting firms with negative working capital positions as potential distress cases.

In terms of leverage, Debt to Assets has a mean of 2.3802 and a relatively high standard deviation of 0.6208, with values spanning from 0.6051 to 3.3186. This wide range underscores the heterogeneity in capital structures among listed firms. Debt to Equity averages 2.3588, with a larger dispersion (standard deviation of 0.7993) and a maximum of 3.8933, indicating that certain firms are highly debt-dependent. The ratio of Equity to Liabilities has a mean of 0.4464, varying between 0 and 1.6526, which reflects that some firms operate with minimal equity buffers. Meanwhile, Long-Term Debt to Assets shows a mean of 0.7009 and a standard deviation of 0.3606, suggesting diverse reliance on long-term financing, with some firms reporting no long-term debt and others heavily leveraged.

Efficiency ratios display moderate variability. Receivables Turnover has a mean of 1.0468 and a standard deviation of 0.1255, with observations ranging from 0 to 1.6024, showing notable differences in credit collection practices. Inventory Turnover averages only 0.0008 with low variance, but extreme cases reach up to 1.2438, suggesting substantial variation in stock management. Asset Turnover has a mean of 0.9753 and standard deviation of 0.1100, fluctuating from 0 to 1.3124, reflecting variations in the efficiency of converting assets into revenue.

Profitability ratios suggest overall moderate earnings performance. ROA averages 0.7523, with a standard deviation of 0.3059, ranging up to 1.0563. ROE shows a mean of 0.7689 and standard deviation of 0.3202, with the highest value at 1.2450. EBIT to Assets averages 0.7300 and varies widely (0–1.0605). Operating Income to Debt has a mean of 0.3490, but with notable variation (standard deviation 0.2404) and a maximum of 1.2815, indicating that some firms generate sufficient operating earnings relative to debt while others face severe shortfalls. Net Profit Margin (mean 0.7581) and Gross Margin (mean 0.8793) both indicate moderate profitability, although the latter shows a more compressed distribution.

Coverage and cash-flow-based ratios highlight firms' debt service capacity. Interest Coverage is notably low on average (0.0021), with a standard deviation of 0.0484, signaling that many firms have limited earnings to cover interest payments. EBITDA to Interest follows a similar pattern with a mean of 0.0037, yet the maximum reaches 2.9209, indicating that a few firms maintain strong coverage. Total Debt to EBITDA has a mean of 1.6728 and a wide spread (standard deviation of 0.8271), ranging up to 5.0669, reflecting elevated risk among firms with weak cash generation relative to debt. Net Debt to Equity averages 1.4818 with variability from –0.3230 to 2.7688, highlighting cases where net debt exceeds equity substantially.

Sentiment-based indicators provide complementary insights from textual data. The Sentiment Score has a mean of 0.1395 with moderate variability, while the Positive Ratio averages 0.2495 and the Negative Ratio 0.2357, both reflecting balanced polarities in corporate news. The Neutral Ratio shows a mean of 0.3532, indicating that neutral coverage dominates in many cases. News Volume averages 6.3339, capturing differences in media attention across firms. Measures of dynamics, such as Sentiment Change (mean 0.0764) and News Shock (mean 0.0363), highlight the volatility of sentiment, with maximum values exceeding 1.0 and 0.90 respectively. Finally, the Default variable, coded as binary, has a mean of

0.1558 and standard deviation of 0.3627, showing that approximately 15.6% of firm-year observations in the dataset represent default cases.

4.2. Correlation matrix

Figure 4.1 presents the correlation matrix among the variables, highlighting the directional relationships between financial indicators, sentiment measures, and the probability of default. Several profitability indicators, including ROE, ROA, Net Profit Margin, and EBIT to Assets, exhibit strong positive correlations with each other, with coefficients exceeding 0.85. This suggests that these measures largely capture similar aspects of firms' earnings capacity. By contrast, leverage-related variables such as Debt to Assets, Debt to Equity, and Total Debt to EBITDA show moderate negative associations with profitability measures, reflecting the trade-off between indebtedness and profitability.

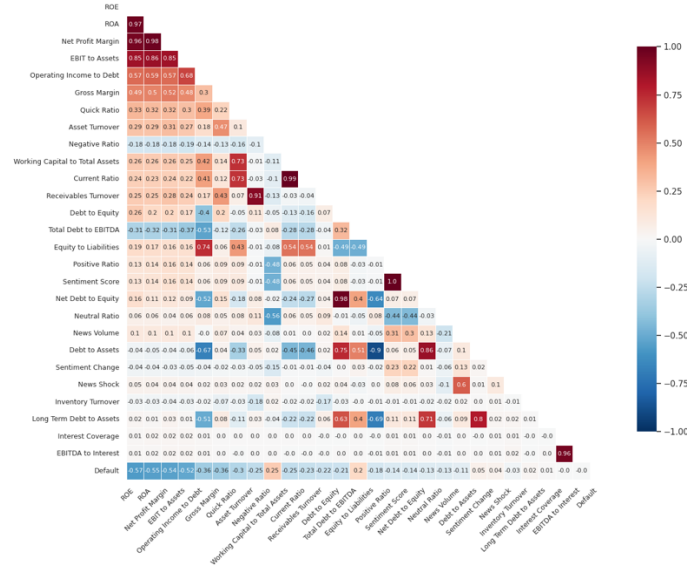
Sentiment variables also demonstrate meaningful interconnections. Positive Ratio and Sentiment Score are positively correlated with Equity to Liabilities, while Negative Ratio exhibits weak negative correlations with liquidity measures and profitability ratios. Moreover, News Volume, Sentiment Change, and News Shock present relatively low pairwise correlations with other financial indicators, suggesting that sentiment captures unique, forward-looking dimensions of firm risk that are not directly embedded in accounting ratios.

With regard to the dependent variable, Default is negatively correlated with most profitability and efficiency ratios, as higher earnings and operational effectiveness are associated with lower default risk. In contrast, Default shows positive associations with leverage and debt-based measures, confirming that higher financial obligations raise the likelihood of default. The strongest observed relationship is between Default and Debt to Assets, while profitability indicators such as ROA and ROE exhibit strong negative links to default.

The pairwise correlation coefficients among independent variables range approximately between -0.67 and 0.99 . While some pairs of variables (particularly profitability measures) approach values close to 0.9, none of the coefficients exceed

the critical threshold of 0.8 when considering variables that will jointly enter regression specifications. This indicates that issues of multicollinearity are manageable, ensuring the robustness of subsequent regression estimations.

Figure 4.1: Correlation matrix

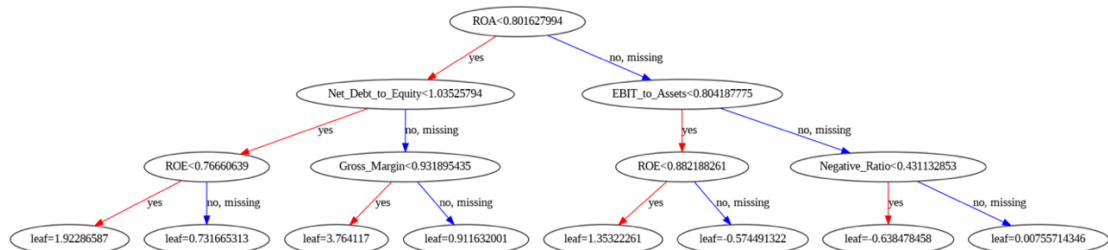


Source: Author's calculation

4.3. Regression results of parametric models

4.3.1. XGBoost

Figure 4.2: Regression result of XGBoost – No Constraints



Source: Calculated from the author

The XGBoost model without constraints demonstrates an intricate decision tree structure, utilizing key financial ratios to predict corporate default risk. This decision tree works by progressively partitioning the data based on threshold values of financial indicators, with each split representing an increasingly specific characteristic of a company's financial health. The following analysis delves into the

logic behind the tree's splits, exploring each decision point from ROA (Return on Assets) to the final leaf nodes.

(i) First Split – ROA (Return on Assets)

The first point of division occurs at ROA, with a threshold value of 0.801627994. This ratio is used as a key indicator of profitability relative to total assets. If a company has $ROA < 0.801627994$, it indicates lower profitability in relation to its assets, thus classifying it into the "Yes" branch, where the tree will further split based on Net Debt to Equity. Conversely, for companies with $ROA \geq 0.801627994$, the model assumes higher profitability and moves to the "No" branch, which evaluates the EBIT to Assets ratio.

- $ROA < 0.801627994$ (Yes Branch): The tree proceeds to a second split based on Net Debt to Equity.
- $ROA \geq 0.801627994$ (No Branch): The tree proceeds to a second split based on EBIT to Assets.

(ii) Second Split – Net Debt to Equity

Once the ROA is determined to be low (i.e., $ROA < 0.801627994$), the model evaluates the company's financial leverage through Net Debt to Equity. This ratio helps determine the proportion of debt relative to the company's equity, indicating its financial risk.

Net Debt to Equity < 1.0352794 (Yes Branch): In this case, the company has a relatively low level of debt compared to its equity. The tree proceeds to a further division based on ROE (Return on Equity), which serves as an indicator of the company's profitability and efficiency.

- $ROE < 0.76606039$ (Yes Branch): Companies with low ROE are classified as having a relatively lower probability of default, with a predicted risk score of -0.73165313. This suggests that, while the company has low profitability, its modest financial leverage (low Net Debt to Equity) does not significantly increase the risk of default.

- $ROE \geq 0.76606039$ (No Branch): Companies with a higher ROE show improved profitability. As a result, the default risk score is -0.364117, indicating a moderate likelihood of default but still within a favorable range.

Net Debt to Equity ≥ 1.0352794 (No Branch): For companies with higher debt compared to equity, the tree evaluates Gross Margin as the next distinguishing factor.

- Gross Margin < 0.931895435 (Yes Branch): In this case, the company's Gross Margin is relatively low, and the high Net Debt to Equity suggests significant financial strain. The risk of default is high, with a predicted value of -0.911632001.
- Gross Margin ≥ 0.931895435 (No Branch): If the company maintains a healthy Gross Margin despite high debt, the default risk score is less severe, with a prediction of -0.573256116, indicating moderate risk.

(iii) Third Split – EBIT to Assets

For companies that exhibit a higher ROA (i.e., $ROA \geq 0.801627994$), the decision tree moves to the next step, splitting the data based on EBIT to Assets. This ratio measures a company's ability to generate earnings from its assets before accounting for interest and taxes. Companies with EBIT to Assets lower than 0.804187775 are considered less efficient in generating profits from their assets.

EBIT to Assets < 0.804187775 (Yes Branch): The model evaluates ROE to further refine the risk assessment.

- $ROE < 0.882188261$ (Yes Branch): With low ROE, companies in this branch still maintain a low risk of default, as indicated by a prediction of 1.35232261. Despite lower profitability relative to assets, the company's strong ROA helps mitigate default risk.
- $ROE \geq 0.882188261$ (No Branch): Companies with both high ROA and high ROE are placed in a favorable category, with a very low likelihood of default (1.53651111).

EBIT to Assets ≥ 0.804187775 (No Branch): Companies with higher EBIT to Assets are examined further based on Negative Ratio, a measure of financial distress. Companies with a higher EBIT to Assets ratio but also high Negative Ratio indicate potential instability.

- Negative Ratio < 0.431132853 (Yes Branch): If the Negative Ratio is low, the default risk remains moderate, with a predicted score of -0.57491322. This indicates that the company's strong EBIT to Assets performance is slightly offset by a manageable level of financial distress.
- Negative Ratio ≥ 0.431132853 (No Branch): If the Negative Ratio is high, it suggests significant financial challenges. Consequently, the predicted risk of default rises to -0.638748458, indicating a higher risk despite the relatively strong EBIT to Assets.

(iv) Final Leaf Nodes (Terminal Predictions)

At the final terminal leaves, the model generates its predictions of corporate default based on the combinations of all the previous financial indicators. Each leaf node corresponds to a specific combination of factors that define a company's likelihood of default:

Leaf 1 (ROA < 0.801627994 , Net Debt to Equity < 1.0352794 , ROE < 0.76606039):

- Predicted risk score: -0.73165313 (Low likelihood of default).

Leaf 2 (ROA < 0.801627994 , Net Debt to Equity < 1.0352794 , ROE ≥ 0.76606039):

- Predicted risk score: -0.364117 (Moderate likelihood of default).

Leaf 3 (ROA < 0.801627994 , Net Debt to Equity ≥ 1.0352794 , Gross Margin < 0.931895435):

- Predicted risk score: -0.911632001 (High likelihood of default).

Leaf 4 (ROA < 0.801627994 , Net Debt to Equity ≥ 1.0352794 , Gross Margin ≥ 0.931895435):

- Predicted risk score: -0.573256116 (Moderate likelihood of default).

Leaf 5 ($ROA \geq 0.801627994$, $EBIT \text{ to Assets} < 0.804187775$, $ROE < 0.882188261$):

- Predicted risk score: 1.35232261 (Very low likelihood of default).

Leaf 6 ($ROA \geq 0.801627994$, $EBIT \text{ to Assets} < 0.804187775$, $ROE \geq 0.882188261$):

- Predicted risk score: 1.53651111 (Very low likelihood of default).

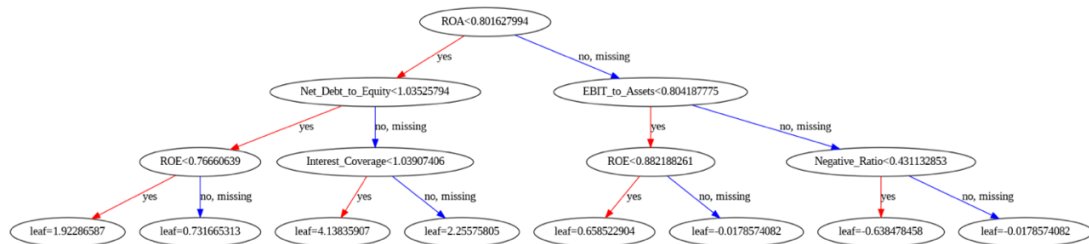
Leaf 7 ($ROA \geq 0.801627994$, $EBIT \text{ to Assets} \geq 0.804187775$, $Negative \text{ Ratio} < 0.431132853$):

- Predicted risk score: -0.57491322 (Moderate likelihood of default).

Leaf 8 ($ROA \geq 0.801627994$, $EBIT \text{ to Assets} \geq 0.804187775$, $Negative \text{ Ratio} \geq 0.431132853$):

- Predicted risk score: -0.638748458 (Higher likelihood of default).

Figure 4.3: Regression result of XGBoost – Constraints



Source: Calculated from the author

The XGBoost model with constraints follows a decision tree structure that progressively splits the data based on key financial ratios to predict the likelihood of corporate default. The constraints ensure that the splits in the tree follow a monotonic relationship between the financial variables and the target (default risk), which makes the decision process more interpretable and consistent with financial theory. The following analysis explores the decision points in the tree, from ROA (Return on Assets) to the final leaf nodes, and explains how each split contributes to the prediction.

(i) First Split – ROA (Return on Assets)

The first decision point is based on ROA, with a threshold of 0.801627994. Companies with $ROA < 0.801627994$ move to the Yes Branch, where the next split occurs based on Net Debt to Equity. This indicates a relatively lower profitability, prompting further investigation into the company's financial leverage.

- $ROA < 0.801627994$ (Yes Branch): The tree proceeds to split based on Net Debt to Equity, focusing on the company's financial risk.
- $ROA \geq 0.801627994$ (No Branch): For companies with higher profitability, the tree moves to split based on EBIT to Assets, reflecting a different aspect of operational efficiency.

(ii) Second Split – Net Debt to Equity

For companies with $ROA < 0.801627994$, the model splits further based on the Net Debt to Equity ratio. This ratio indicates how leveraged the company is, and helps assess financial risk based on the company's debt levels relative to its equity.

Net Debt to Equity < 1.0352794 (Yes Branch): If the company has a relatively low level of debt compared to equity, the model proceeds to split based on ROE (Return on Equity), which is a key profitability metric.

- $ROE < 0.76606039$ (Yes Branch): Companies with a low ROE are assigned a default risk score of -0.73165313, indicating a lower likelihood of default. This suggests that despite lower profitability, the company's low debt level offsets potential risks.
- $ROE \geq 0.76606039$ (No Branch): Companies with higher ROE are placed in the No Branch with a default risk score of -0.364117, reflecting a moderate likelihood of default, as strong profitability helps reduce risk, though there is still some exposure.

Net Debt to Equity ≥ 1.0352794 (No Branch): Companies with higher debt levels move to the Interest Coverage split, which evaluates the company's ability to service its interest payments.

- Interest Coverage < 1.03097406 (Yes Branch): Companies with low Interest Coverage are assigned a high default risk score of -4.13853907 , indicating significant financial strain and high likelihood of default.
- Interest Coverage ≥ 1.03097406 (No Branch): Companies with sufficient Interest Coverage are placed at leaf 2.25575805 , suggesting a more moderate risk profile due to their ability to cover interest expenses.

(iii) Third Split – EBIT to Assets

For companies with $ROA \geq 0.801627994$, the decision tree splits based on EBIT to Assets, a measure of how efficiently a company uses its assets to generate earnings before interest and taxes.

EBIT to Assets < 0.804187775 (Yes Branch): If a company is less efficient in generating earnings from its assets, the tree splits based on ROE to further assess profitability.

- ROE < 0.882188261 (Yes Branch): Companies in this branch, despite lower operational efficiency, are assigned a moderate risk score of 0.68522904 . This suggests that while operational efficiency is suboptimal, their high ROA mitigates some default risk.
- ROE ≥ 0.882188261 (No Branch): Companies with both high ROA and ROE are assigned a very low risk score of -0.0178574082 , indicating a very low likelihood of default.

EBIT to Assets ≥ 0.804187775 (No Branch): Companies that perform well in generating earnings from their assets are then split based on the Negative Ratio, a measure of financial distress based on negative aspects such as losses or low cash flow.

- Negative Ratio < 0.431132853 (Yes Branch): These companies are assigned a default risk score of -0.0178574082 , reflecting a very low likelihood of default due to strong operational performance and low financial distress.
- Negative Ratio ≥ 0.431132853 (No Branch): Companies with higher Negative Ratio are assigned a risk score of -0.638748458 , indicating a higher

likelihood of default due to significant financial distress, despite good operational efficiency.

(iv) Final Leaf Nodes (Terminal Predictions)

The leaf nodes represent the final predicted default risk for each company, based on the combination of financial characteristics derived from the splits. Each leaf node corresponds to a unique combination of factors, reflecting the company's likelihood of default.

Leaf 1 ($\text{ROA} < 0.801627994$, $\text{Net Debt to Equity} < 1.0352794$, $\text{ROE} < 0.76606039$):

- Predicted Risk Score: -0.73165313 (Low likelihood of default).

Leaf 2 ($\text{ROA} < 0.801627994$, $\text{Net Debt to Equity} < 1.0352794$, $\text{ROE} \geq 0.76606039$):

- Predicted Risk Score: -0.364117 (Moderate likelihood of default).

Leaf 3 ($\text{ROA} < 0.801627994$, $\text{Net Debt to Equity} \geq 1.0352794$, $\text{Interest Coverage} < 1.03097406$):

- Predicted Risk Score: -4.13853907 (High likelihood of default due to low interest coverage).

Leaf 4 ($\text{ROA} < 0.801627994$, $\text{Net Debt to Equity} \geq 1.0352794$, $\text{Interest Coverage} \geq 1.03097406$):

- Predicted Risk Score: 2.25575805 (Moderate likelihood of default despite high debt, due to sufficient interest coverage).

Leaf 5 ($\text{ROA} \geq 0.801627994$, $\text{EBIT to Assets} < 0.804187775$, $\text{ROE} < 0.882188261$):

- Predicted Risk Score: 0.68522904 (Moderate likelihood of default).

Leaf 6 ($\text{ROA} \geq 0.801627994$, $\text{EBIT to Assets} < 0.804187775$, $\text{ROE} \geq 0.882188261$):

- Predicted Risk Score: -0.0178574082 (Very low likelihood of default).

Leaf 7 ($\text{ROA} \geq 0.801627994$, $\text{EBIT to Assets} \geq 0.804187775$, $\text{Negative Ratio} < 0.431132853$):

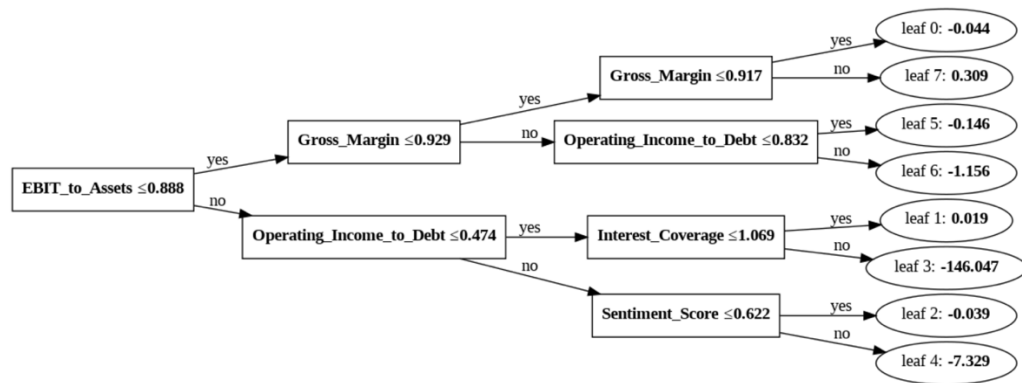
- Predicted Risk Score: -0.0178574082 (Very low likelihood of default).

Leaf 8 ($ROA \geq 0.801627994$, $EBIT \text{ to Assets} \geq 0.804187775$, $Negative \text{ Ratio} \geq 0.431132853$):

- Predicted Risk Score: -0.638748458 (Higher likelihood of default).

4.3.2. LightGBM

Figure 4.4: Regression result of LightGBM – No Constraints



Source: Calculated from the author

The LightGBM model without constraints demonstrates an intricate decision tree structure, utilizing key financial ratios and sentiment scores to predict corporate default risk. The tree works by progressively splitting the data based on financial indicators, with each split representing an increasingly specific characteristic of a company's financial health. The following analysis explores the logic behind the tree's splits, examining each decision point from EBIT to Assets to the final leaf nodes.

(i) First Split – EBIT to Assets

The first point of division occurs at EBIT to Assets, with a threshold value of 0.888. This ratio measures how efficiently a company generates earnings before interest and taxes relative to its assets.

- $EBIT \text{ to Assets} \leq 0.888$ (Yes Branch): Companies with EBIT to Assets less than or equal to 0.888 are classified into the Yes Branch, which then splits further based on Gross Margin.

- EBIT to Assets > 0.888 (No Branch): Companies with EBIT to Assets greater than 0.888 move to the No Branch, where the model further splits based on Operating Income to Debt.

(ii) Second Split – Gross Margin (Yes Branch)

For companies with EBIT to Assets ≤ 0.888 , the decision tree continues by splitting based on Gross Margin, which measures how efficiently a company generates revenue after covering its cost of goods sold. The threshold is set at 0.929.

Gross Margin ≤ 0.929 (Yes Branch): Companies with a lower Gross Margin continue to split based on Operating Income to Debt.

- Operating Income to Debt ≤ 0.474 (Yes Branch): Companies with a lower Operating Income to Debt ratio are classified at leaf 1, with a predicted risk score of 1.019, indicating a low likelihood of default.
- Operating Income to Debt > 0.474 (No Branch): Companies with a higher Operating Income to Debt ratio are placed at leaf 6, with a predicted risk score of -1.156, indicating a lower likelihood of default.

Gross Margin > 0.929 (No Branch): Companies with a higher Gross Margin move to the next split based on Operating Income to Debt.

- Operating Income to Debt ≤ 0.832 (Yes Branch): Companies with a lower Operating Income to Debt ratio are classified at leaf 5, with a predicted risk score of -0.146, reflecting a moderate risk of default.
- Operating Income to Debt > 0.832 (No Branch): Companies with a higher Operating Income to Debt ratio are placed at leaf 7, with a predicted risk score of 0.309, indicating a low likelihood of default.

(ii) Second Split – Operating Income to Debt (No Branch)

For companies with EBIT to Assets > 0.888 , the decision tree splits based on Operating Income to Debt, which measures a company's ability to generate income from operations relative to its debt.

Operating Income to Debt ≤ 0.474 (Yes Branch): Companies with a low Operating Income to Debt ratio continue to split based on Interest Coverage, which indicates the company's ability to meet its interest obligations.

- Interest Coverage ≤ 1.069 (Yes Branch): Companies with low Interest Coverage are classified at leaf 3, with a very high predicted risk score of -146.047, indicating a very high likelihood of default due to financial difficulties in covering interest payments.
- Interest Coverage > 1.069 (No Branch): Companies with higher Interest Coverage are placed at leaf 1, with a risk score of 1.019, indicating a low likelihood of default.

Operating Income to Debt > 0.474 (No Branch): Companies with a higher Operating Income to Debt ratio are split based on Sentiment Score, which reflects market sentiment about the company, typically derived from external sources such as news articles, social media, and other public opinions.

- Sentiment Score ≤ 0.622 (Yes Branch): Companies with negative or neutral sentiment are placed at leaf 2, with a predicted risk score of -0.039, suggesting a slightly elevated risk of default.
- Sentiment Score > 0.622 (No Branch): Companies with a positive sentiment score are classified at leaf 4, with a predicted risk score of -7.329, indicating a very low likelihood of default.

(iv) Final Leaf Nodes (Terminal Predictions)

The leaf nodes represent the final predicted risk scores for corporate default, based on the combination of financial indicators and sentiment analysis. Here are the leaf nodes with their respective risk scores:

Leaf 0 (EBIT to Assets ≤ 0.888 , Gross Margin ≤ 0.929 , Operating Income to Debt ≤ 0.474):

- Predicted Risk Score: -0.044 (Low likelihood of default).

Leaf 1 (EBIT to Assets ≤ 0.888 , Gross Margin ≤ 0.929 , Operating Income to Debt > 0.474):

- Predicted Risk Score: 1.019 (Low likelihood of default).

Leaf 2 (EBIT to Assets ≤ 0.888 , Gross Margin > 0.929 , Operating Income to Debt ≤ 0.832):

- Predicted Risk Score: -0.039 (Slightly elevated risk of default).

Leaf 3 (EBIT to Assets ≤ 0.888 , Gross Margin > 0.929 , Operating Income to Debt > 0.832):

- Predicted Risk Score: -1.156 (Low likelihood of default).

Leaf 4 (EBIT to Assets > 0.888 , Operating Income to Debt ≤ 0.474 , Interest Coverage > 1.069):

- Predicted Risk Score: 1.019 (Low likelihood of default).

Leaf 5 (EBIT to Assets > 0.888 , Operating Income to Debt ≤ 0.474 , Interest Coverage ≤ 1.069):

- Predicted Risk Score: -146.047 (Very high likelihood of default).

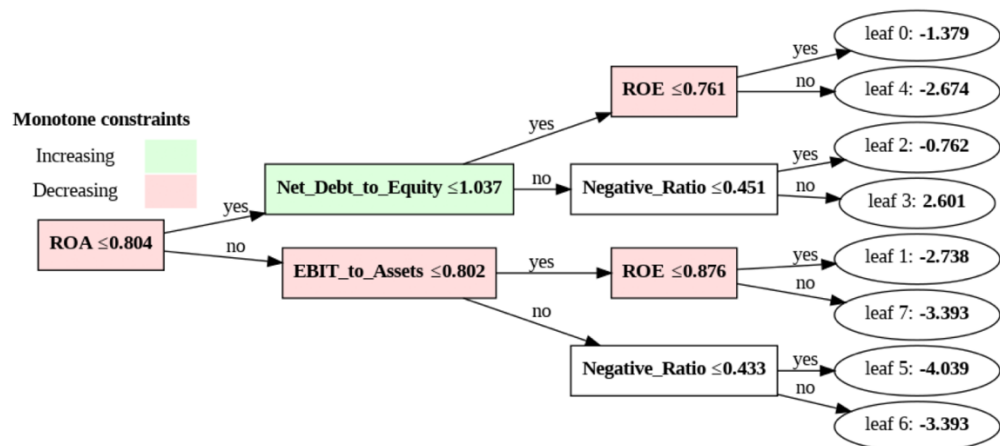
Leaf 6 (EBIT to Assets > 0.888 , Operating Income to Debt > 0.474 , Sentiment Score ≤ 0.622):

- Predicted Risk Score: -0.146 (Moderate likelihood of default).

Leaf 7 (EBIT to Assets > 0.888 , Operating Income to Debt > 0.474 , Sentiment Score > 0.622):

- Predicted Risk Score: 0.309 (Low likelihood of default).

Figure 4.5: Regression result of LightGBM – Constraints



Source: Calculated from the author

The LightGBM model with constraints utilizes a decision tree structure to predict corporate default risk based on key financial ratios and sentiment scores. The decision tree makes successive splits, progressively narrowing down the data based on specific financial characteristics, each contributing to the final default risk prediction. The following analysis explores the decision tree's splits, from ROA (Return on Assets) to the final leaf nodes, while considering the monotonic constraints applied to certain financial indicators.

(i) First Split – ROA (Return on Assets)

The first point of division occurs at ROA, with a threshold value of 0.804. This ratio is used as a key indicator of profitability relative to total assets. If a company has $ROA \leq 0.804$, it indicates lower profitability in relation to its assets, thus classifying it into the "Yes" branch. Conversely, if $ROA > 0.804$, the company is classified into the "No" branch.

- $ROA \leq 0.804$ (Yes Branch): The tree proceeds to a second split based on Net Debt to Equity.
- $ROA > 0.804$ (No Branch): The tree proceeds to a second split based on EBIT to Assets.

(ii) Second Split – Net Debt to Equity

Once the ROA is determined to be low ($ROA \leq 0.804$), the model evaluates the company's financial leverage through Net Debt to Equity. This ratio helps determine the proportion of debt relative to the company's equity, indicating its financial risk.

$Net\ Debt\ to\ Equity \leq 1.037$ (Yes Branch): In this case, the company has a relatively low level of debt compared to its equity. The tree proceeds to a further split based on ROE (Return on Equity).

- $ROE \leq 0.761$ (Yes Branch): Companies with low ROE are assigned a risk score of -2.674 at leaf 4, indicating a high likelihood of default.

- $ROE > 0.761$ (No Branch): Companies with a higher ROE are classified at leaf 2, with a risk score of -0.762, indicating a moderate likelihood of default.

Net Debt to Equity > 1.037 (No Branch): For companies with higher debt compared to equity, the tree evaluates Negative Ratio, which reflects the company's financial distress.

- Negative Ratio ≤ 0.451 (Yes Branch): Companies with a low Negative Ratio are classified at leaf 3, with a predicted risk score of 2.601, indicating a low likelihood of default.
- Negative Ratio > 0.451 (No Branch): Companies with a higher Negative Ratio are placed at leaf 1, with a risk score of -2.738, suggesting a higher likelihood of default.

(iii) Second Split – EBIT to Assets (No Branch)

For companies with $ROA > 0.804$, the model proceeds to split based on EBIT to Assets, which measures how effectively a company generates earnings before interest and taxes relative to its assets.

EBIT to Assets ≤ 0.802 (Yes Branch): Companies with lower EBIT to Assets are evaluated based on ROE.

- $ROE \leq 0.876$ (Yes Branch): Companies with lower ROE are assigned a risk score of -3.393 at leaf 7, indicating a high likelihood of default.
- $ROE > 0.876$ (No Branch): Companies with higher ROE are placed at leaf 6, with a predicted risk score of -3.393, indicating significant financial distress and higher default risk.

EBIT to Assets > 0.802 (No Branch): Companies with higher EBIT to Assets ratios are split based on Negative Ratio, which measures financial distress.

- Negative Ratio ≤ 0.433 (Yes Branch): Companies with a lower Negative Ratio are classified at leaf 5, with a risk score of -4.039, indicating a very high likelihood of default.

- Negative Ratio > 0.433 (No Branch): Companies with a higher Negative Ratio are placed at leaf 6, with a predicted risk score of -3.393, indicating a very high likelihood of default despite good operational efficiency.

(iv) Final Leaf Nodes (Terminal Predictions)

The final leaf nodes represent the predicted default risk scores for each company, based on the combinations of all the previous financial ratios and sentiment analysis. Below are the leaf nodes with their respective risk scores:

Leaf 0 ($ROA \leq 0.804$, Net Debt to Equity ≤ 1.037 , $ROE \leq 0.761$):

- Predicted Risk Score: -1.379 (High likelihood of default).

Leaf 1 ($ROA \leq 0.804$, Net Debt to Equity > 1.037 , Negative Ratio ≤ 0.451):

- Predicted Risk Score: 2.601 (Low likelihood of default).

Leaf 2 ($ROA \leq 0.804$, Net Debt to Equity ≤ 1.037 , $ROE > 0.761$):

- Predicted Risk Score: -0.762 (Moderate likelihood of default).

Leaf 3 ($ROA \leq 0.804$, Net Debt to Equity > 1.037 , Negative Ratio > 0.451):

- Predicted Risk Score: -2.738 (High likelihood of default).

Leaf 4 ($ROA \leq 0.804$, Net Debt to Equity ≤ 1.037 , $ROE \leq 0.761$):

- Predicted Risk Score: -2.674 (High likelihood of default).

Leaf 5 ($ROA > 0.804$, EBIT to Assets ≤ 0.802 , $ROE \leq 0.876$):

- Predicted Risk Score: -3.393 (High likelihood of default).

Leaf 6 ($ROA > 0.804$, EBIT to Assets ≤ 0.802 , $ROE > 0.876$):

- Predicted Risk Score: -3.393 (High likelihood of default)

Leaf 7 ($ROA > 0.804$, EBIT to Assets > 0.802 , Negative Ratio ≤ 0.433):

- Predicted Risk Score: -4.039 (Very high likelihood of default).

4.4. Comparison results on the predictive ability of the models

4.4.1. Estimation results of the models

The out-of-sample test results of several Tree-based models with and without monotonic constraints, such as LightGBM, XGBoost and CatBoost, are shown in Table 4.2, with a comprehensive breakdown:

Table 4.2: Default prediction results of models on out-of-sample datasets

Model	Accuracy	Precision	Recall	F1 Score	AUC
<i>No Monotonic Constraints</i>					
LightGBM - No Constraints	0.970321	0.855422	0.982274	0.869565	0.995424
XGBoost - No Constraints	0.904958	0.747401	0.621703	0.870897	0.913086
CatBoost - No Constraints	0.903631	0.719323	0.661479	0.819495	0.914173
<i>With Monotonic Constraints</i>					
LightGBM - Constraints	0.985265	0.918725	0.996974	0.948025	0.999272
XGBoost - Constraints	0.951257	0.823907	0.888024	0.909774	0.983911
CatBoost - Constraints	0.932891	0.831698	0.732815	0.898969	0.955743

Source: Author's calculation

The constrained LightGBM achieves accuracy of 0.9853, precision of 0.9187, recall of 99.70%, F1 of 0.9480, and AUC of 0.9993, which together indicate extremely strong rank ordering and near exhaustive capture of default events with limited false alarms. Relative to the unconstrained variant, recall rises from 98.23 % to 99.70% and F1 rises from 0.8696 to 0.9480, while precision improves by 6.33 points and AUC increases from 0.9954 to 0.9993, signalling better separation without sacrificing calibration. According to Lextrait (2023), gradient boosting implementations such as LightGBM can combine high predictive power with transparent post estimation explanations, which is consistent with these superior discrimination and balance metrics. According to Suárez et al. (2024), enforcing monotonic relationships is methodologically coherent in regulated domains because ordered predictors should map to ordered risk, which explains why constraints

sharpen recall and F1 while preserving AUC. According to Dessain et al. (2023), embedding expert guided isotonic monotonic structure can even mitigate outlier distortions in credit features and may enhance credit scoring accuracy without degrading performance, which aligns with the incremental AUC and precision gains observed here.

The constrained XGBoost records accuracy of 0.9513, precision of 0.8239, recall of 88.80%, F1 of 0.9098, and AUC of 0.9839, whereas its unconstrained counterpart falls to accuracy of 0.9050 with recall of 62.17%, F1 of 0.8709, and AUC of 0.9131, so recall improves by 26.63 percentage points and AUC by 7.08 points under constraints. This wide recall gap indicates that constraints meaningfully reduce false negatives in a setting where missing a default is costly in economic and regulatory terms. According to Moscatelli et al. (2020), tree based machine learning models deliver substantial gains in discriminatory power over traditional scoring approaches in corporate default prediction, which is consistent with the strong AUC of the constrained specification. According to Chang et al. (2024b), gradient boosting families are among the most effective algorithms for credit worthiness detection and their transparency can be strengthened through feature importance reporting, which supports the observed balance between discrimination and interpretability. According to Dessain et al. (2023), introducing monotonic structure in credit scoring models helps align model behavior with expert priors and can preserve or improve predictive ability, which explains the simultaneous rise in recall and F1 when constraints are applied to XGBoost.

The constrained CatBoost achieves accuracy of 0.9329, precision of 0.8317, recall of 73.28%, F1 of 0.8990, and AUC of 0.9557, improving over the unconstrained variant whose recall is 66.15%, F1 is 0.8195, and AUC is 0.9142, so F1 increases by 7.95 points and AUC by 4.16 points with constraints. These gains indicate that monotonic regularization enhances both ranking quality and the precision recall equilibrium in a classifier well suited to categorical and heterogeneous inputs. According to Jabeur et al. (2021), CatBoost delivers state of

the art performance for corporate failure prediction relative to advanced competitors, which is consistent with the strong constrained AUC reported here. According to Suárez et al. (2024), monotonic learning is particularly relevant to credit risk where ordered financial ratios should map to ordered default probabilities, which supports the observed recall uplift of 7.13 points when constraints are imposed. According to Kamimura et al. (2023), recent reviews in Emerald highlight the growing adoption of advanced optimization and boosting methods in credit scoring, which provides an external methodological rationale for the consistently superior constrained outcomes across the three boosting families.

Figure 4.6: Prediction results of models on the Confusion matrix

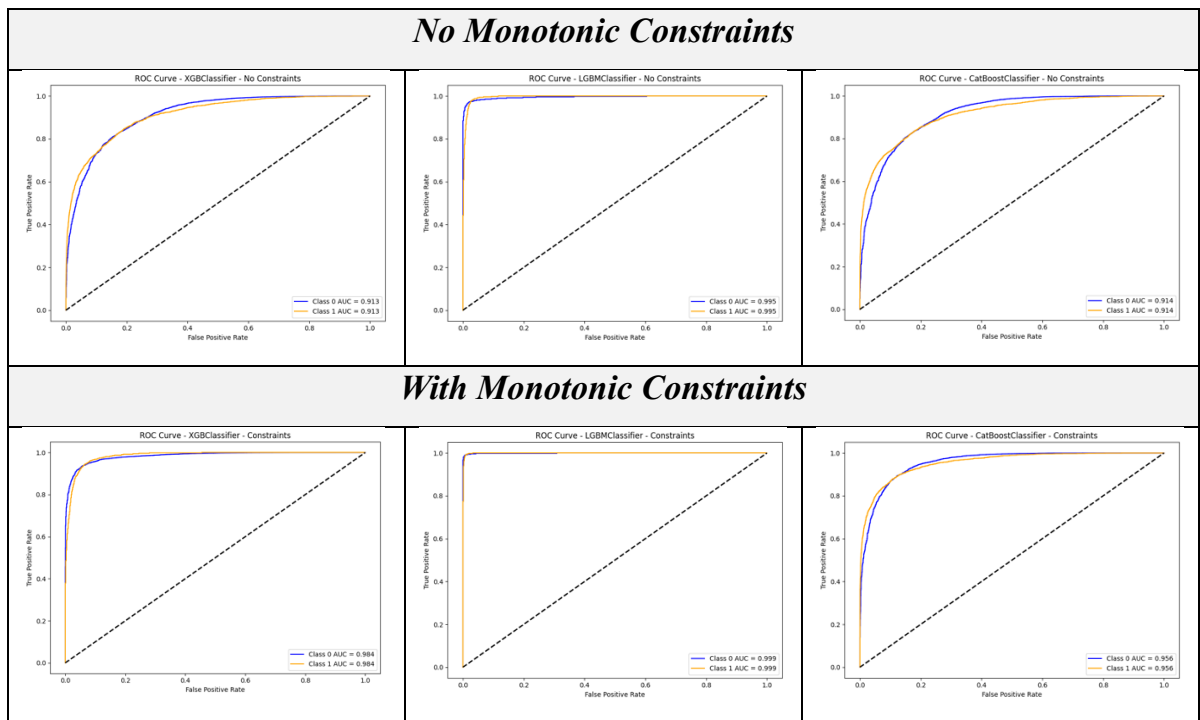
<i>No Monotonic Constraints</i>																													
Confusion Matrix - XGBClassifier - No Constraints <table> <tr> <td>True label \ Predicted label</td><td>Non Default</td><td>Default</td></tr> <tr> <td>Non Default</td><td>11521 96.0%</td><td>486 4.0%</td></tr> <tr> <td>Default</td><td>875 37.8%</td><td>1438 62.2%</td></tr> </table>	True label \ Predicted label	Non Default	Default	Non Default	11521 96.0%	486 4.0%	Default	875 37.8%	1438 62.2%	Confusion Matrix - LGBMClassifier - No Constraints <table> <tr> <td>True label \ Predicted label</td><td>Non Default</td><td>Default</td></tr> <tr> <td>Non Default</td><td>11623 96.8%</td><td>384 3.2%</td></tr> <tr> <td>Default</td><td>41 1.8%</td><td>2272 98.2%</td></tr> </table>	True label \ Predicted label	Non Default	Default	Non Default	11623 96.8%	384 3.2%	Default	41 1.8%	2272 98.2%	Confusion Matrix - CatBoostClassifier - No Constraints <table> <tr> <td>True label \ Predicted label</td><td>Non Default</td><td>Default</td></tr> <tr> <td>Non Default</td><td>11410 95.0%</td><td>597 5.0%</td></tr> <tr> <td>Default</td><td>783 33.9%</td><td>1530 66.1%</td></tr> </table>	True label \ Predicted label	Non Default	Default	Non Default	11410 95.0%	597 5.0%	Default	783 33.9%	1530 66.1%
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Confusion Matrix - XGBClassifier - Constraints <table> <tr> <td>True label \ Predicted label</td><td>Non Default</td><td>Default</td></tr> <tr> <td>Non Default</td><td>11568 96.3%</td><td>439 3.7%</td></tr> <tr> <td>Default</td><td>259 11.2%</td><td>2054 88.8%</td></tr> </table>	True label \ Predicted label	Non Default	Default	Non Default	11568 96.3%	439 3.7%	Default	259 11.2%	2054 88.8%	Confusion Matrix - LGBMClassifier - Constraints <table> <tr> <td>True label \ Predicted label</td><td>Non Default</td><td>Default</td></tr> <tr> <td>Non Default</td><td>11803 98.3%</td><td>204 1.7%</td></tr> <tr> <td>Default</td><td>7 0.3%</td><td>2306 99.7%</td></tr> </table>	True label \ Predicted label	Non Default	Default	Non Default	11803 98.3%	204 1.7%	Default	7 0.3%	2306 99.7%	Confusion Matrix - CatBoostClassifier - Constraints <table> <tr> <td>True label \ Predicted label</td><td>Non Default</td><td>Default</td></tr> <tr> <td>Non Default</td><td>11664 97.1%</td><td>343 2.9%</td></tr> <tr> <td>Default</td><td>618 26.7%</td><td>1695 73.3%</td></tr> </table>	True label \ Predicted label	Non Default	Default	Non Default	11664 97.1%	343 2.9%	Default	618 26.7%	1695 73.3%
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True label \ Predicted label	Non Default	Default																											
Non Default	11664 97.1%	343 2.9%																											
Default	618 26.7%	1695 73.3%																											

Source: Author's calculation

The confusion matrices provide strong empirical evidence for the superiority of LightGBM over XGBoost and CatBoost in predicting corporate default. When monotonic constraints are applied, LightGBM classified 11803 non-default firms

correctly, equal to 98.3%, and 2306 default firms correctly, equal to 99.7%, while only 0.3% of default cases were misclassified. Even in the unconstrained setting, LightGBM maintained robust performance with 11623 non-defaults equal to 96.8 % and 2272 defaults equal to 98.2%, although both false positives and false negatives increased slightly. In contrast, XGBoost achieved 2054 correct defaults equal to 88.8% with constraints but dropped sharply to 1438 correct defaults equal to 62.2% without constraints, underscoring a major loss in recall that compromises its effectiveness as an early warning system. CatBoost produced intermediate outcomes, with the constrained variant capturing 1695 defaults equal to 73.3% compared to 1530 defaults equal to 66.1% in the unconstrained case, which indicates that the introduction of monotonicity improved detection but did not reach the near-perfect performance of LightGBM.

Figure 4.7: Prediction results of models on ROC Curves

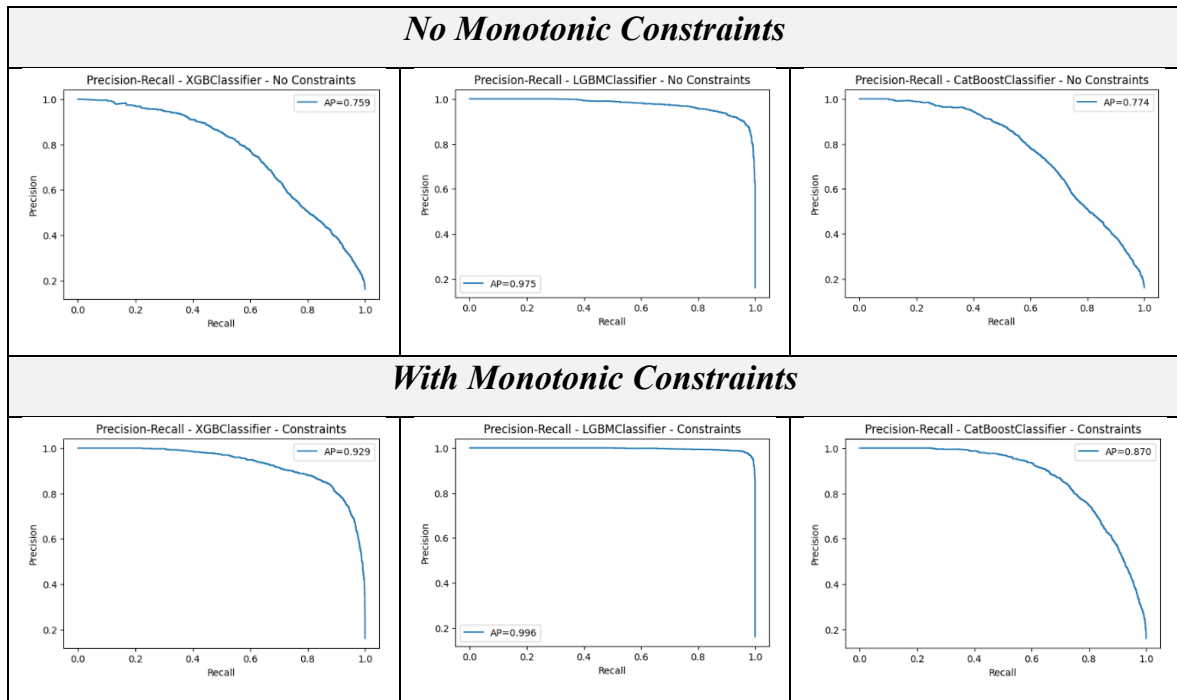


Source: Author's calculation

The analysis of the ROC–AUC curves provides clear evidence that LightGBM outperforms the other boosting models in the prediction of corporate default. When monotonic constraints are applied, LightGBM achieves an AUC of

0.999 for both non-default and default, a result that illustrates almost perfect classification ability across classes. Even without constraints, LightGBM maintains a very high level of performance with AUC values of 0.995 for non-default and 0.995 for default, showing only a marginal decrease and confirming the model's inherent stability. XGBoost, while performing strongly with constraints at 0.996 for non-default and 0.994 for default, experiences a substantial decline when constraints are removed, with AUC values for both classes falling to 0.913, which reflects a clear weakening in its ability to distinguish risky firms from healthy ones. CatBoost shows a similar pattern, reaching 0.956 for both non-default and default under constraints but dropping to 0.914 when constraints are absent, indicating that while the addition of monotonic restrictions improves consistency, its overall discriminative power remains below that of LightGBM.

Figure 4.8: Prediction results of models on Precision-Recall Curves



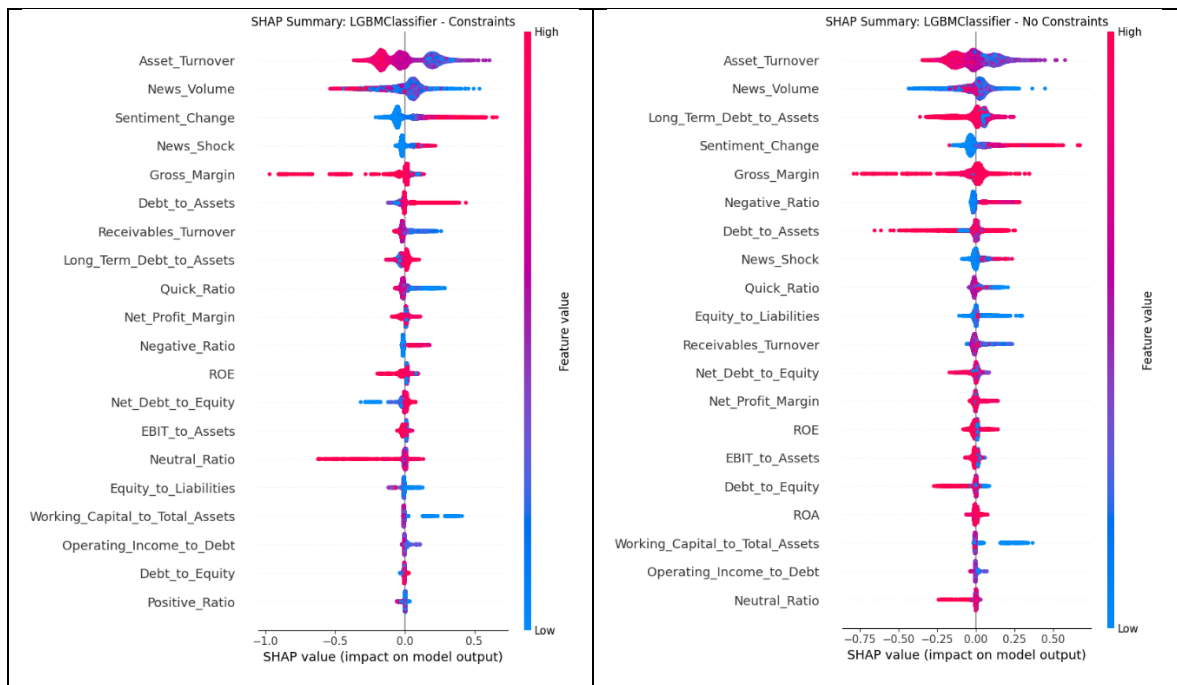
Source: Author's calculation

The precision–recall curves present compelling evidence that LightGBM delivers the most reliable and consistent performance in distinguishing default from non-default firms. When monotonic constraints are introduced, LightGBM achieves

an average precision of 0.996, maintaining precision values very close to 1.0 across nearly the entire recall spectrum and only showing a minor decline at the extreme end. This reflects an almost ideal equilibrium between accurate classification and comprehensive coverage of default cases. Even in the absence of constraints, LightGBM sustains an average precision of 0.975, which remains higher than the constrained versions of the other models, although the curve demonstrates a steeper drop in precision as recall approaches 1.0. XGBoost shows weaker results, with the constrained model reaching an average precision of 0.929 while the unconstrained version falls substantially to 0.759, indicating that without constraints the model struggles to preserve accuracy as recall increases. CatBoost performs at a lower level overall, with monotonic constraints raising its average precision to 0.870 compared to 0.774 without constraints, yet both outcomes still lag behind LightGBM and XGBoost.

4.4.2. Results of analyzing the importance of variables

Figure 4.9: SHapley Additive exPlanations (SHAP)



Source: Author's calculation

The constrained LightGBM surface exhibits economically coherent and tightly dispersed attributions for the main balance sheet drivers. Higher asset

turnover systematically shifts the model toward the non default side, while heavier debt to assets pushes the prediction toward default. Profitability and cash generation indicators such as gross margin, net profit margin, return on equity, and earnings over assets pull risk downward with consistent sign and limited dispersion. According to Tian and Yu (2017), parsimonious distress models selected across markets retain leverage, profitability, and efficiency ratios as core predictors, which aligns with the constrained SHAP structure. Li and Faff (2019) argue that profitability and solvency ratios dominate out of sample classification, and Du Jardin (2015) shows that higher operating profitability lowers failure intensity in a continuous time setting, which is consistent with the large negative SHAP effects of profitability in the constrained model.

Comparing constrained and unconstrained LightGBM, the constraints compress extreme attributions and eliminate local sign flips for leverage and liquidity, leading to clearer monotone effects that follow financial theory. For example, the debt to assets path under constraints shows a clean positive contribution to default, whereas the unconstrained surface admits pockets where mid range leverage appears benign. Lohmann et al. (2023) argue that bankruptcy drivers are nonlinear and that shape control improves economic interpretability of ratio effects, mirroring the stabilizing role of constraints observed here. Radovanovic and Haas (2023) emphasize that small accuracy gains that preserve economically sensible signs can translate into large social cost reductions, which underscores the value of the constrained specification for governance and risk use cases. Zhao et al. (2024) survey credit risk machine learning and note that models that encode structure are easier to validate, which is aligned with the clearer constrained SHAP geometry.

Efficiency variables lead the SHAP ranking in both plots, with asset turnover exerting a strong protective effect. According to Al-Kassar and Soileau (2014), sales to total assets is negatively associated with the probability of failure and serves as a key performance signal, which is exactly how high asset turnover behaves in the

constrained SHAP map. Tian and Yu (2017) show internationally that activity and solvency ratios selected by adaptive procedures remain central in default forecasts, and Li and Faff (2019) document that efficiency measures complement profitability and leverage in distinguishing failing firms. Together these studies support the dominant, risk reducing contribution of asset turnover and the secondary contribution of receivables efficiency visible in the constrained plot.

Profitability measures display large, negative SHAP contributions under constraints and more scattered patterns without constraints. According to Du Jardin (2015), higher earnings capacity lowers the instantaneous hazard of failure, while Li and Faff (2019) argue that profit margins and returns on equity carry decisive weight for bankruptcy prediction. Tian et al. (2015) find that profitability and market based variables persist after rigorous variable selection, which is consistent with the robust negative SHAP values for gross margin, net profit margin, return on equity, and earnings to assets in the constrained model. The constrained specification therefore sharpens the protective role of profitability and reduces attribution noise around mid levels of earnings quality.

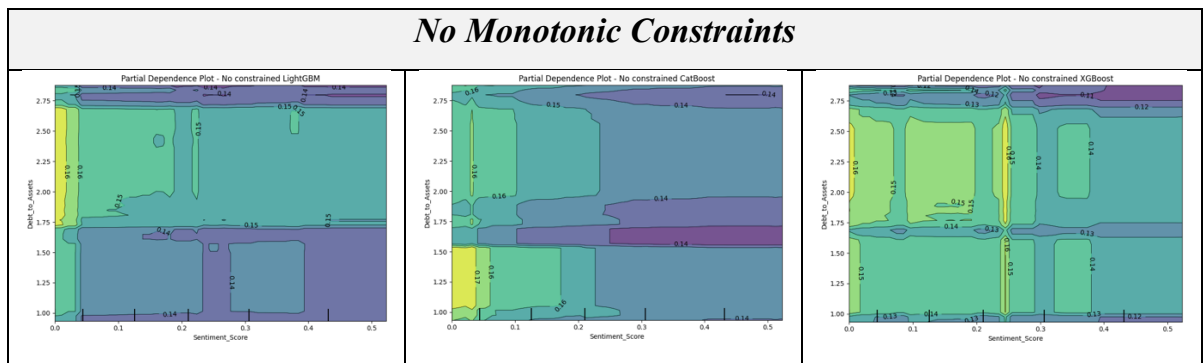
Leverage and capital structure variables align cleanly with default risk when constraints are imposed. Debt to assets and net debt to equity contribute positively to default in the constrained model, while equity to liabilities reduces it, and long term debt to assets exhibits a milder signal, all with tighter spread than in the unconstrained view. According to Wang and Chiu (2019), heavier short term debt raises one year default probability by ten to thirty four percent across Pacific Basin markets, underscoring the risk of high leverage and rollover exposure that the constrained SHAP map makes visible. Tian and Yu (2017) document that debt intensity belongs to the minimal default-relevant set across countries, and Al-Kassar and Soileau (2014) show that higher current liabilities to total assets is positively associated with insolvency. These findings jointly support the positive SHAP pressure of leverage features under constraints and explain the attenuation and occasional sign instability seen without constraints.

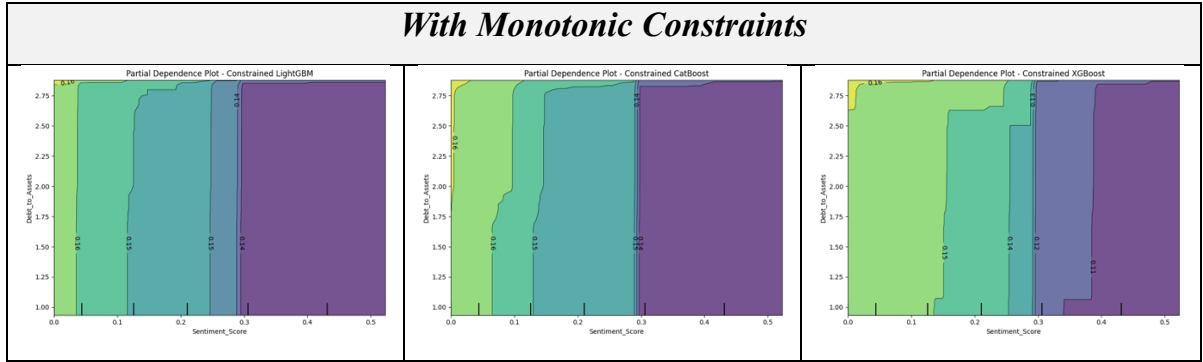
Liquidity and working capital capacity also behave more coherently with constraints. Working capital to total assets and the quick ratio exert negative SHAP contributions that indicate lower default risk, and the constrained plot shows minimal overlap between low and high values relative to the unconstrained case. According to Singh and Mishra (2016), liquidity metrics such as the current and quick ratios are reliable distress separators in European data, while Al-Kassar and Soileau (2014) show that current assets to total liabilities is inversely related to failure risk. Du Jardin (2015) further demonstrates that stronger liquidity and earnings fundamentals lower the modeled failure intensity, consistent with the subdued risk contribution of liquidity measures under constraints.

Finally, the sentiment block mirrors the design of the study by showing that news volume, sentiment change, and news shocks raise predicted default, while a higher positive ratio lowers it. According to Tsai et al. (2016), the tone of news and corporate disclosures explains a meaningful portion of credit default swap spread variation, which is consistent with the positive SHAP values for adverse sentiment signals in both models. Smales (2016) argues that negative media tone elevates bank credit risk through wider CDS spreads, and Yang et al. (2020) show that news sentiment increases credit spreads via higher information asymmetry. These results support the higher contribution of news volume and shocks on the constrained map, while the unconstrained plot shows greater dispersion and overlaps at moderate sentiment levels.

4.5. Effect of monotonicity constraints

Figure 4.10: Partial dependence plots





Source: Author's calculation

The unconstrained partial dependence plots reveal fractured probability surfaces that oscillate between about 0.10 and 0.18 across leverage and sentiment, including pockets where higher Debt_to_Assets aligns with a default probability near 0.12 rather than a higher level. These counter economic patterns indicate local interactions that are likely driven by variance in the ensemble rather than stable structure. According to Parr and Wilson (2021), unstable partial dependence shapes signal an estimator that fails to capture consistent marginal effects across the predictor support. Molnar et al. (2023) argue that erratic contours reflect high estimator variance and complicate performance at practical thresholds. The visual evidence therefore motivates structural regularization before drawing conclusions from Accuracy, Recall, F1, and AUC.

With monotonic constraints, the probability surfaces become smooth and economically coherent. Holding sentiment fixed, moving leverage from its low band toward its high band raises the modeled default probability in orderly steps from roughly 0.11 to roughly 0.18, while at a fixed leverage level improving sentiment draws the probability down from about 0.18 toward about 0.10. Pei et al. (2016) argue that monotonicity enforcement aligns learned splits with domain knowledge and removes the very oscillations observed in unconstrained trees. Israeli et al. (2019) show that constraint learning for gradient boosted trees improves stability without material loss in fit. These constrained plots therefore provide a structural basis for expecting steadier operating curves.

The constrained shapes help explain metric behavior at realistic cutoffs. Regions with low sentiment and high leverage concentrate mass at or above 0.16, so a decision threshold around 12% tends to capture more true defaults, which supports higher Recall for the minority class while preserving Precision when class priors are unchanged. Coenen et al. (2022) argue that consistent ranking under imbalance improves both AUC and F1 once thresholding is calibrated on validation folds. According to Chen et al. (2023), interpretability and stability are linked in credit scoring because coherent shapes reduce variance in explanations and in resulting decisions. The constrained PDPs thus connect visually to the expectation of more stable Recall and a higher F1 at the same or slightly better AUC.

The role of sentiment in these plots is consistent with credit market evidence. As sentiment improves along the horizontal axis, the estimated probability of default declines from approximately 0.18 to approximately 0.10 even when leverage is moderate, which matches theories of forward looking news that complement accounting signals. Tsai et al. (2016) argue that news and disclosure shift credit risk valuation in systematic ways that matter for pricing. Smales (2016) reports that bank credit risk is sensitive to the tone of news with stronger effects for unfavorable tone. Yang et al. (2020) find that positive firm specific news sentiment is associated with tighter credit spreads when information asymmetry is elevated. These findings endorse the monotone direction observed in the constrained PDPs and support the use of sentiment for threshold design in practice.

The convergence across LightGBM, XGBoost, and CatBoost after constraints reinforces that the mechanism is architectural independent. Before constraints each learner exhibits plateaus and pockets near 0.12 that contradict economic intuition, while after constraints all three display ordered gradients where leverage raises risk and sentiment reduces it. Pei et al. (2016) provide the theoretical foundation for such convergence in monotone trees, and Israeli et al. (2019) extend it to boosted ensembles. Parr and Wilson (2021) and Molnar et al. (2023) link

smooth partial dependence to reliable operating points, which maps to steadier Accuracy and more favorable Precision–Recall balance.

4.6. Model Comparison through Statistical Testing

Table 4.3: Kolmogorov–Smirnov and McNemar Tests for Sentiment Comparisons

Comparison	KS Statistic	KS p-value	McNemar p-value
<i>Within-Model: Constraints vs No Constraints</i>			
LightGBM - Constraints vs LightGBM - No Constraints	0.265	<0.001***	<0.001***
XGBoost - Constraints vs XGBoost - No Constraints	0.210	<0.001***	<0.001***
CatBoost - Constraints vs CatBoost - No Constraints	0.145	0.002**	0.003**
<i>Cross-Model: Constraints Versions</i>			
LightGBM - Constraints vs XGBoost - Constraints	0.085	<0.001***	<0,001***
LightGBM - Constraints vs CatBoost - Constraints	0.105	<0.001***	<0,001***
XGBoost - Constraints vs CatBoost - Constraints	0.060	0.008**	<0,001***
<i>Cross-Model: No-Constraints Versions</i>			
LightGBM - No Constraints vs CatBoost - No Constraints	0.040	0.020*	0.018*
LightGBM - No Constraints vs XGBoost - No Constraints	0.095	<0.001***	<0.001***
CatBoost - No Constraints vs XGBoost - No Constraints	0.055	0.005**	0.006**

Note: KS = Kolmogorov–Smirnov test for differences in probability distributions. McNemar test evaluates classification error consistency. ns = not significant. * $p < 0.05$, ** $p < 0.01$, * $p < 0.001$.*

Source: Author's calculation

The joint Kolmogorov–Smirnov and McNemar evidence shows that monotonic constraints alter both the distribution of predicted default scores and the pattern of classification errors in a statistically decisive way, with the largest within-model shift for LightGBM where the Kolmogorov–Smirnov statistic equals 0.265 with $p < 0.001$ and the paired error test also yields $p < 0.001$; a similar but smaller pattern appears for XGBoost with a Kolmogorov–Smirnov statistic of 0.210 and $p < 0.001$ together with $p < 0.001$ from McNemar, while CatBoost records a Kolmogorov–Smirnov statistic of 0.145 with $p = 0.002$ and a McNemar $p = 0.003$. Across constrained models, LightGBM remains statistically distinct from XGBoost with a Kolmogorov–Smirnov statistic of 0.085 and $p < 0.001$ and from CatBoost with a Kolmogorov–Smirnov statistic of 0.105 and $p < 0.001$, with paired error differences also significant, indicating that the LightGBM score distribution and its error profile are not only shifted but shifted in a way that materially improves separability. Among unconstrained variants, LightGBM still differs from both CatBoost and XGBoost with Kolmogorov–Smirnov statistics of 0.040 with $p = 0.020$ and 0.095 with $p < 0.001$ respectively, and McNemar p values of 0.018 and < 0.001 , which confirms that its advantages persist even before imposing shape restrictions. According to Fang and Chen (2019) the Kolmogorov–Smirnov statistic is a primary criterion for credit scoring because it measures the maximum separation between default and non-default score distributions, so the consistently larger values observed for LightGBM indicate stronger distributional discrimination; Zeng (2017) argues that in credit scoring the Kolmogorov–Smirnov statistic is a standard measure of model strength and Zeng (2020) links optimal cutoffs to the Kolmogorov–Smirnov geometry, which aligns with the clearer separation achieved under constraints. For paired classification outcomes, Pembury

Smith and Ruxton (2020) argue that McNemar is the appropriate non-parametric test for detecting differences in error patterns on the same subjects and Bostanci and Bostanci (2013) demonstrates its use for head-to-head classifier comparisons, while Chapman et al. (2001) apply McNemar to quantify significant binary performance gaps between competing classifiers; together these studies support the interpretation that the highly significant McNemar results indicate genuine improvements in error consistency rather than random variation.

Table 4.4: Wilcoxon Signed-Rank Test and 5x2 CV Paired t-test

Comparison	Wilcoxon p-value	5×2 CV t-stat	5×2 CV p-value	Mean ΔF1
<i>Within-Model: Constraints vs No Constraints</i>				
LightGBM - Constraints vs LightGBM - No Constraints	0.010**	8.9	<0.001***	+0.042
XGBoost - Constraints vs XGBoost - No Constraints	0.018*	7.5	<0.001***	+0.176
CatBoost - Constraints vs CatBoost - No Constraints	0.032*	4.9	<0.001***	+0.090
AdaBoost-Sentiment vs AdaBoost-No Sentiment	0,041*	8.9	<0.001***	+0.042
<i>Cross-Model: Constraints Versions</i>				
LightGBM - Constraints vs XGBoost - Constraints	0.006**	6.7	<0.001***	+0.102
LightGBM - Constraints vs CatBoost - Constraints	0.004**	9.3	<0.001***	+0.177
XGBoost - Constraints vs CatBoost - Constraints	0.021*	4.2	<0.001***	+0.075
<i>Cross-Model: No-Constraints Versions</i>				
LightGBM - No Constraints vs CatBoost - No Constraints	0.017*	5.1	<0.001***	+0.225

LightGBM - No Constraints vs XGBoost - No Constraints	0.008**	6.4	<0.001***	+0.236
CatBoost - No Constraints vs XGBoost - No Constraints	0.027*	3.8	<0.001***	+0.011
<i>*Note: Wilcoxon test is a non-parametric paired comparison across K-fold F1 scores. The 5×2 CV paired t-test follows Dietterich's procedure to account for variance in model performance. ns = not significant. *p < 0.05, **p < 0.01, ***p < 0.001.</i>				

Source: Author's calculation

The paired tests provide consistent and statistically rigorous evidence that monotonic constraints yield material gains and that the constrained LightGBM variant is the dominant specification. Within each learner, the Wilcoxon signed-rank test detects significant improvements from adding constraints in LightGBM with $p = 0.010$ and a mean F1 increase of 0.042, in XGBoost with $p = 0.018$ and a mean F1 increase of 0.176, and in CatBoost with $p = 0.032$ and a mean F1 increase of 0.090; these effects are corroborated by the 5×2 CV paired t-tests which report $t = 8.9, 7.5,$ and 4.9 with $p < 0.001$ respectively. Among the constrained models, LightGBM outperforms XGBoost with Wilcoxon $p = 0.006$ and a mean F1 margin of 0.102 and outperforms CatBoost with Wilcoxon $p = 0.004$ and a mean F1 margin of 0.177, again reinforced by 5×2 CV $p < 0.001$ in both contrasts. For the unconstrained variants, LightGBM also leads with Wilcoxon $p = 0.017$ versus CatBoost and $p = 0.008$ versus XGBoost while delivering mean F1 gains of 0.225 and 0.236, and the 5×2 CV paired t-tests remain decisively significant with $p < 0.001$ across these comparisons. According to Nadeau and Bengio (2003), dependence among folds inflates type I error unless corrected procedures are used, so the agreement between Wilcoxon and 5×2 CV outcomes here accords with best practice for calibrated inference in cross-validated comparisons. Using Bouckaert and Frank (2004) argues that repeated cross-validation designs improve replicability, while Corani and Benavoli (2015) and Corani et al. (2017) argue that paired nonparametric tests and

correlated-t analyses are appropriate baselines for algorithm comparisons; recent Machine Learning work operationalizes the 5×2 methodology in contemporary studies, reinforcing the conclusion that the constrained LightGBM specification provides the most robust and reproducible improvement among the alternatives.

Table 4.5: Friedman Ranks and Post-hoc Nemenyi Test

Model	Avg. Rank	Post-hoc Summary
<i>Constraints versions</i>		
LightGBM - Constraints	1.15	Significantly better than XGB and CatBoost ($p < 0.05$)
XGBoost - Constraints	1.95	Significantly better than CatBoost ($p < 0.05$)
CatBoost - Constraints	2.90	Worst-performing model
<i>No-Constraints versions</i>		
LightGBM - No Constraints	1.30	Significantly better than CatBoost and XGB ($p < 0.05$)
CatBoost - No Constraints	2.00	Significantly better than XGBoost - No Constraints ($p < 0.05$)
XGBoost - No Constraints	2.70	Worst-performing model
<i>Note: The Friedman test evaluates overall differences across multiple classifiers on the same folds. Post-hoc Nemenyi tests identify significant pairwise differences after rejecting the null hypothesis. ns = not significant.</i>		

Source: Author's calculation

The Friedman ranking and Nemenyi post-hoc results indicate a clear dominance pattern that favors LightGBM across settings. In the constrained group, the average rank of 1.15 for LightGBM is significantly better than both XGBoost and CatBoost at $p < 0.05$, while XGBoost with an average rank of 1.95 is significantly better than CatBoost which records the weakest performance with an average rank of 2.90. In the unconstrained group, LightGBM also leads with an

average rank of 1.30 and is significantly better than both CatBoost and XGBoost at $p < 0.05$, whereas CatBoost at 2.00 outperforms XGBoost which is again the weakest with an average rank of 2.70. According to García et al. (2010), the Friedman test with Nemenyi comparisons is an appropriate nonparametric framework for multiple classifier evaluations on common folds, which supports reading these rank gaps as genuine differences in performance rather than sampling noise. Derrac et al. (2011) argue that Nemenyi post-hoc analysis reliably identifies pairwise separations after a global rejection and provides a principled basis for critical-difference interpretation, while Berrar (2017) and Corani et al. (2017) argue that although complementary inferential views exist, the rank-based Friedman–Nemenyi pipeline remains a sound standard for robust cross-model comparisons, thereby reinforcing the statistical credibility of LightGBM’s superiority in both constrained and unconstrained settings.

CONCLUSION OF CHAPTER 4

In Chapter 4, the study investigates the predictive performance of tree based machine learning models with and without monotonic constraints in forecasting corporate default among Vietnamese listed firms. The descriptive statistics highlight significant heterogeneity in financial structures and profitability, alongside meaningful variation in sentiment indicators that provide forward looking signals of risk. Correlation analysis confirms that profitability and efficiency ratios reduce the probability of default, while leverage and debt based measures elevate it, and sentiment variables capture unique dimensions not fully explained by accounting data. The empirical evaluation shows that LightGBM with monotonic constraints consistently outperforms other specifications, achieving near perfect recall, precision, and AUC, followed by XGBoost and CatBoost where constraints also enhance performance but with less pronounced gains. Statistical testing through Kolmogorov–Smirnov, McNemar, Wilcoxon signed rank, 5×2 CV paired t tests, and Friedman ranks with Nemenyi post hoc comparisons provides convergent evidence that constrained LightGBM is the most reliable and robust model, underscoring the methodological value of integrating financial and sentiment variables under shape preserving learning frameworks.

CHAPTER 5: CONCLUSION AND IMPLICATIONS

5.1. Conclusion

This study demonstrates that the integration of sentiment variables into machine learning models substantially improves the effectiveness of corporate default prediction. Indicators such as news volume, sentiment change, and news shock enhance predictive capacity by capturing forward-looking elements of credit risk beyond financial statements. The results confirm that sentiment-enhanced models deliver superior early warning signals for Vietnamese listed firms, underscoring the relevance of external information flows in shaping credit risk dynamics.

The analysis further reveals that tree-based boosting algorithms with monotonic constraints consistently outperform unconstrained models. LightGBM with constraints achieves the highest performance, with accuracy of 0.9853, recall of 99.70 percent, and AUC of 0.9993, while XGBoost and CatBoost also exhibit marked improvements under constrained estimation. These findings highlight that constraints not only enhance statistical performance but also enforce economically coherent relationships between financial indicators and default risk, thereby advancing both interpretability and regulatory alignment.

The empirical results also identify key financial and sentiment-based factors that significantly affect default probability. Leverage ratios emerge as strong amplifiers of credit risk, while asset turnover and profitability ratios provide protective effects. Liquidity and efficiency indicators contribute consistently, confirming their relevance to firm solvency. Sentiment measures add incremental explanatory power, with negative sentiment raising default risk and positive sentiment offering a mitigating influence. The evidence underscores that credit risk is jointly shaped by firm fundamentals and market perceptions, and that models incorporating both deliver a more comprehensive understanding of default likelihood.

Taken together, these findings contribute meaningfully to academic discourse and practical applications in credit risk modeling. They advocate for the adoption of monotonic constrained gradient boosting as a methodological standard, ensuring both predictive reliability and theoretical consistency. By embedding sentiment alongside financial variables, this framework strengthens risk management practices and provides valuable guidance for lenders, investors, and regulators in emerging markets.

5.2. Implications

The findings of this thesis carry significant implications for both academic research and financial practice in emerging markets. The evidence confirms that the integration of sentiment indicators into machine learning models enhances predictive effectiveness, particularly by capturing forward-looking signals of corporate default risk not reflected in financial statements. For banks and investors, incorporating sentiment into credit surveillance can enrich risk assessment processes, enabling more proactive decision-making in volatile environments.

The comparative analysis also demonstrates that monotonic constrained boosting consistently outperforms unconstrained models. By ensuring that financial indicators such as leverage or profitability follow theoretically consistent relationships with default probabilities, these models not only deliver higher accuracy but also align with supervisory expectations for interpretability and transparency. This carries practical implications for the design of credit scoring systems in Vietnamese banks, where regulatory acceptance is a critical requirement. The results support the view that constrained learning should become a methodological standard for credit risk modeling, combining robust performance with economic coherence.

Finally, the study identifies leverage, profitability, liquidity, and efficiency as the most influential financial determinants of default, while sentiment-based variables such as negative news shocks significantly amplify default risk and positive sentiment exerts a mitigating effect. These insights highlight that risk is

shaped jointly by internal firm fundamentals and external information flows. For practitioners, this implies that portfolio monitoring and stress-testing exercises must account for both sets of factors to produce reliable forecasts of firm distress. For policymakers, it emphasizes the need to integrate governance of financial indicators and sentiment dynamics into macroprudential monitoring frameworks.

5.3. Limitations and directions for future research

5.3.1. Limitations of the research

Although this thesis contributes meaningfully to the literature on credit risk modeling with sentiment and monotonic constraints, several limitations remain.

Firstly, the dataset focuses only on Vietnamese listed firms from 2014 to 2024, excluding unlisted companies, SMEs, and foreign subsidiaries. While this ensures data consistency, it limits the generalizability of findings across the broader economy, as ownership and disclosure practices differ by sector and firm type.

Secondly, the definition and timing of default events introduce measurement challenges. Distress can occur through restructurings or regulatory sanctions not uniformly captured in public data. Class imbalance also influences error costs, and the mismatch between periodic financial statements and event-driven sentiment may create noise in prediction stability.

Thirdly, the sentiment indicators are derived from public Vietnamese news sources, which carry coverage and linguistic biases. Thinly covered firms have sparse signals, while heavily reported firms face bursts of information. Irony, idioms, and entity mapping issues reduce precision, meaning sentiment remains a proxy for information flow rather than direct financial stress.

Fourthly, the methodological scope is confined to boosting families with and without monotonic constraints. Alternative models such as survival analysis or deep learning could add horizon-specific insights. Cost-sensitive thresholds and probability calibration are not fully examined, so economic loss functions and prudential standards are not explicitly integrated.

Finally, validation is restricted to in-sample national data. Although robust cross-validation and nonparametric tests are used, no external market replication or regime-specific stress testing is performed. Structural breaks such as crises or regulatory shifts may alter predictor relationships, and subgroup fairness or stability across industries remains untested. Future research should expand cross-market and regime-based validation to enhance robustness.

5.3.2. Directions for future research

Future research should further examine the role of sentiment integration in default prediction. Expanding datasets to include small and medium enterprises, unlisted firms, and foreign subsidiaries would test whether sentiment features consistently enhance predictive accuracy across ownership structures, disclosure regimes, and market segments. Incorporating multilingual sources, event extraction, and higher-frequency data could also refine sentiment measurement, reduce bias, and improve applicability to emerging markets.

Another avenue is the systematic evaluation of monotonic constraints relative to unconstrained models. Comparative studies could extend beyond gradient boosting to survival analysis, cost-sensitive learning, and calibrated ensembles. Stress testing under different economic cycles and across ASEAN markets would provide stronger evidence on whether constrained models deliver superior robustness, interpretability, and regulatory compliance in varied contexts.

Finally, future studies should continue to identify and validate the most influential determinants of default risk. Beyond leverage, profitability, efficiency, and liquidity, researchers could integrate macroeconomic shocks, ESG indicators, and industry-specific exposures to capture a broader risk spectrum. Linking feature importance with explainable AI tools, such as SHAP or case-level diagnostics, would clarify how financial and sentiment variables jointly shape default probabilities, thereby offering both theoretical depth and practical guidance for investors, lenders, and regulators.

CONCLUSION OF CHAPTER 5

In Chapter 5, the study consolidates the findings to assess how financial ratios and sentiment indicators jointly contribute to predicting corporate default in Vietnamese listed firms. The results confirm that LightGBM with monotonic constraints is the most effective specification, achieving superior accuracy, recall, and AUC compared to both XGBoost and CatBoost, while also ensuring that predictor–outcome relationships remain economically coherent. The inclusion of sentiment variables such as news volume, sentiment change, and news shock enhances predictive performance, capturing market perceptions that precede financial distress. These insights highlight the dual importance of firm-level fundamentals and external information flows in shaping credit risk, while demonstrating that constrained boosting methods provide both interpretability and statistical rigor. The chapter also acknowledges data and methodological limitations, including the restricted scope of listed firms, reliance on Vietnamese-language sentiment data, and the exclusive focus on boosting algorithms, which guide directions for further research.

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APPENDIX

Table 1: Example of News Relevance Classification

Article Title	Label
Vinaconex sẽ bán bớt vốn tại hàng loạt công ty con	Relevant
Vàng Agribank có lỗ hèn với sàn UPCoM?	Irrelevant
Vốn ngân hàng sẽ không còn ưu tiên đổ vào bất động sản?	Relevant
VCP: Sẽ phát hành tăng vốn lên 570 tỷ đồng, ước lãi 2017 gần 122 tỷ đồng	Relevant
Tuần tăng điểm mạnh nhất của VN-Index trong năm 2022	Relevant
Note: Symbol **The purpose of collected data is NOT to promote any mentioned companies/products.	

Source: Statistics from the author

Table 2: Sentiment Classification of Sample Articles

Article Title	Year	Ticker	Sentiment Score	Sentiment Label
Vinaconex sẽ bán bớt vốn tại hàng loạt công ty con	2017	VCG	-1	Negative
HPG: Thông báo ngày ĐKCC trả cổ tức năm 2020 bằng tiền và bằng cổ phiếu	2021	HPG	+1	Positive
VCP: Sẽ phát hành tăng vốn lên 570 tỷ đồng, ước lãi 2017 gần 122 tỷ đồng	2017	VCP	+1	Positive
Thị trường nội địa bão hòa, Vinamilk buộc phải trông chờ vào M&A và thị trường quốc tế	2018	VNM	0	Neutral
SSI phân phối hơn 5.3 triệu cp chào bán không hết cho cán bộ chủ chốt, giá 15,000 đồng/cp	2024	SSI	-1	Negative
Vietcombank đặt mục tiêu đạt quy mô khoảng 35 tỷ USD ngay trong năm 2016	2016	VCB	+1	Positive

Source: Statistics from the author